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Article

Novel pyrrolopyridone bromodomain and extra-terminal motif (BET) inhibitors effective in endocrine-resistant ER+ breast cancer with acquired resistance to fulvestrant and palbociclib

Yangfeng Li, Jiong Zhao, Lauren Gutgesell, Zhengnan Shen, Kiira Ratia, Katherine Dye, Oleksii Dubrovyskyii, Huiping Zhao, Fei Huang, Debra A. Tonetti, Gregory RJ Thatcher, and Rui Xiong

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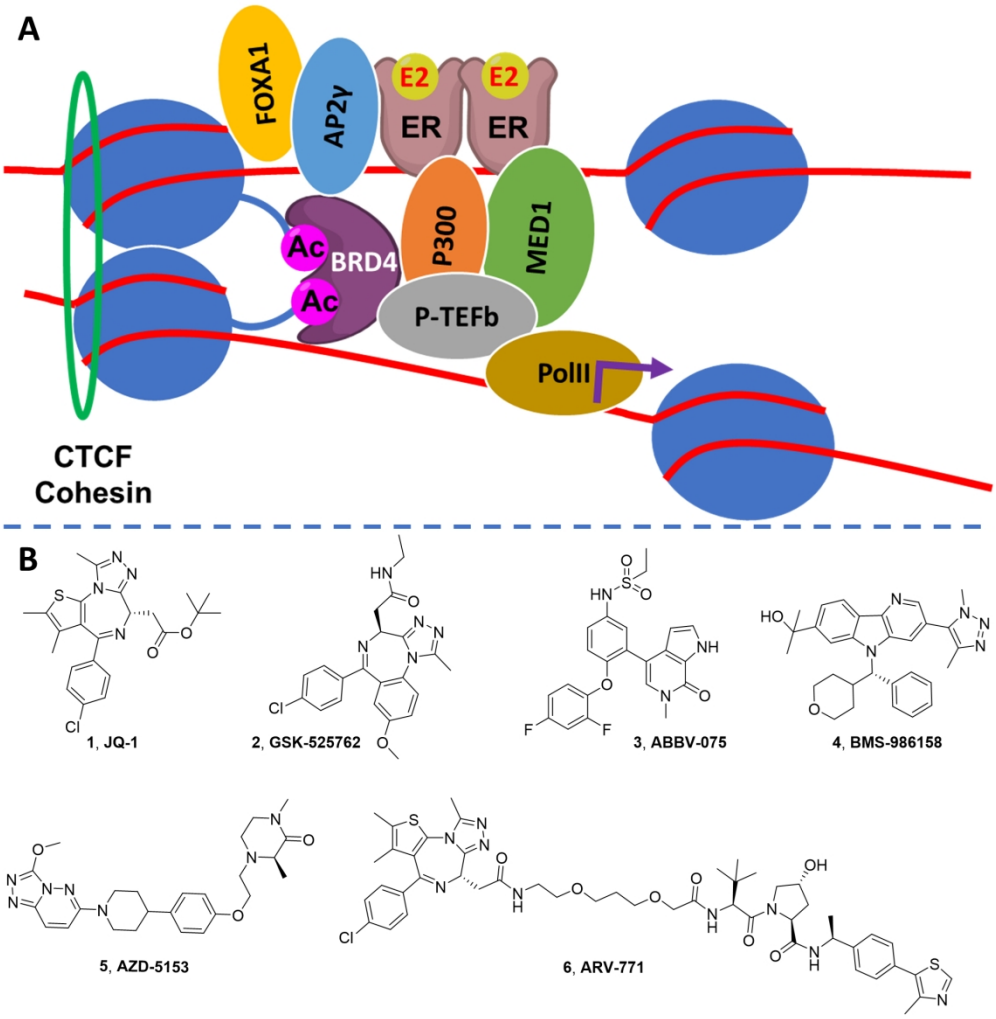

Figure 1. BRD4 is a coregulator of ER signaling in breast cancer. A) ER+ breast cancer is a transcriptionally-driven cancer: E2-liganded ER localizes to ERE forming a transcriptional complex, including proteins such as MED1, P300 and FOXA1; BRD4 recruitment to this complex at acetylated histones and cohesin recruitment, causes transcriptional elongation via P-TEFb and RNA polymerase II. Targeting the ER transcriptional complex through BRD4 provides an alternative approach either as monotherapy or in combination with SERDs to treat ER+ breast cancer; B) Structures of benchmark BET inhibitors.

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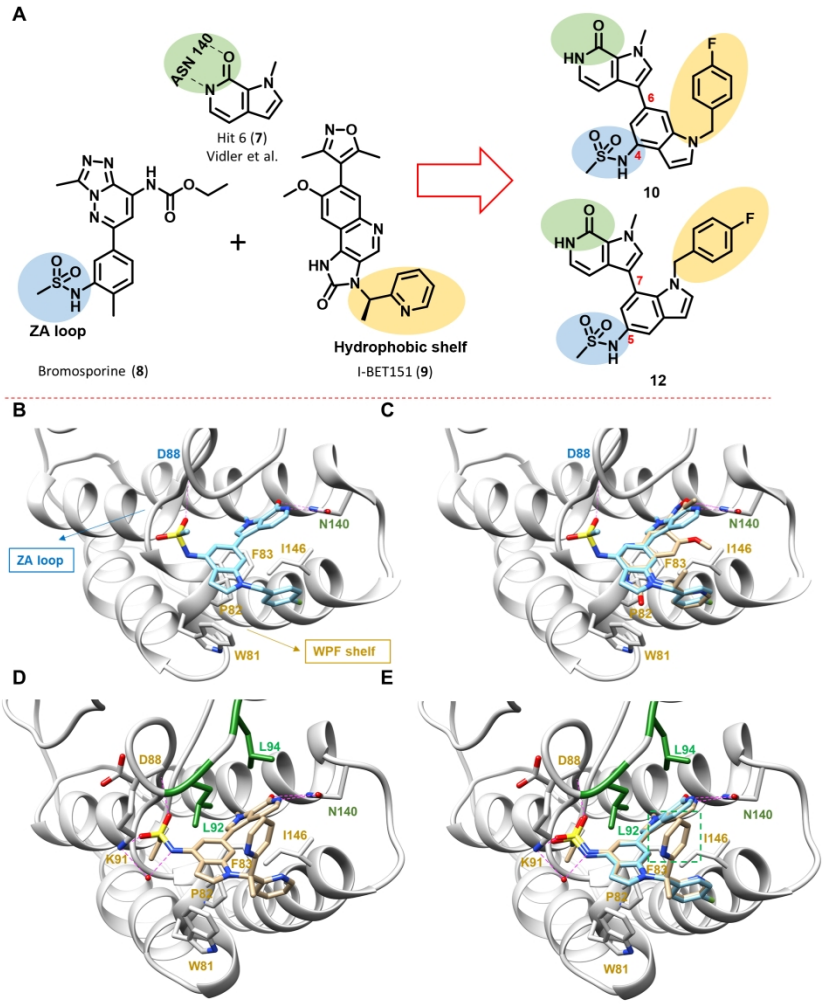

Figure 2. Design of pyrrolopyridone-based BET inhibitors. A) Design concept inspired by earlier reported BET inhibitors: 1) pyrrolopyridone, 7, identified from a virtual screen forms bidentate hydrogen-bonding interaction with Asn140, highlighted in green, 2) sulfonamide group of 8 interacts with ZA loop via a putative hydrogen-bonding interaction, highlighted in blue; and 3) the heteroaryl moiety of 9 interacts with the hydrophobic WPF shelf, highlighted in yellow. Combining these design parameters leads to the design of 10 and 12 as novel BET inhibitors; B) Predicted binding pose of compound 10 binding to BRD4-BD1 using Glide (pdb: 3ZYU): key interacting residues from docking are highlighted and labeled; C) Overlay of the predicted binding pose of 10 (blue) with 9 (beige) from pdb: 3ZYU. D) New co-crystal of compound 27 bound to BRD4-BD1, showing four hydrogen bonds with N140, D88 and K91, and one additional water-mediated hydrogen bond with K91. The "leucine clamp" (L92 and L94) is highlighted in green and other key interacting residues are highlighted and labeled in red. E) Overlay of the predicted binding pose of 10 (blue) with the experimentally confirmed binding pose of 27 (beige) clearly showing the second pyridine ring of 27 (boxed in green) that forms a hydrophobic interaction with L92 and L94.

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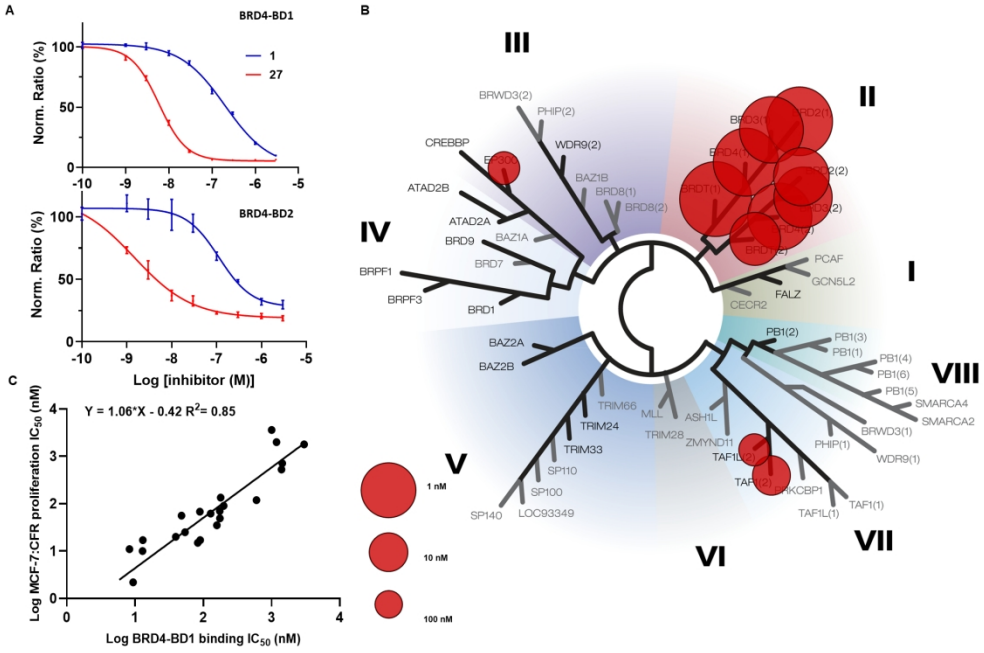

Figure 3. Binding to BRD4-BD1 correlates with cellular antiproliferative potency. A) BRD4-BD1 and BRD4-BD2 binding measured by TR-FRET for 1 compared to 27 (data normalized to DMSO as 100%); B) BromoScan of compound 27 against 40 bromodomain proteins; C) Correlation between BRD4-BD1 binding (TR-FRET) assay and potency for growth inhibition of MCF-7:CFR cells.

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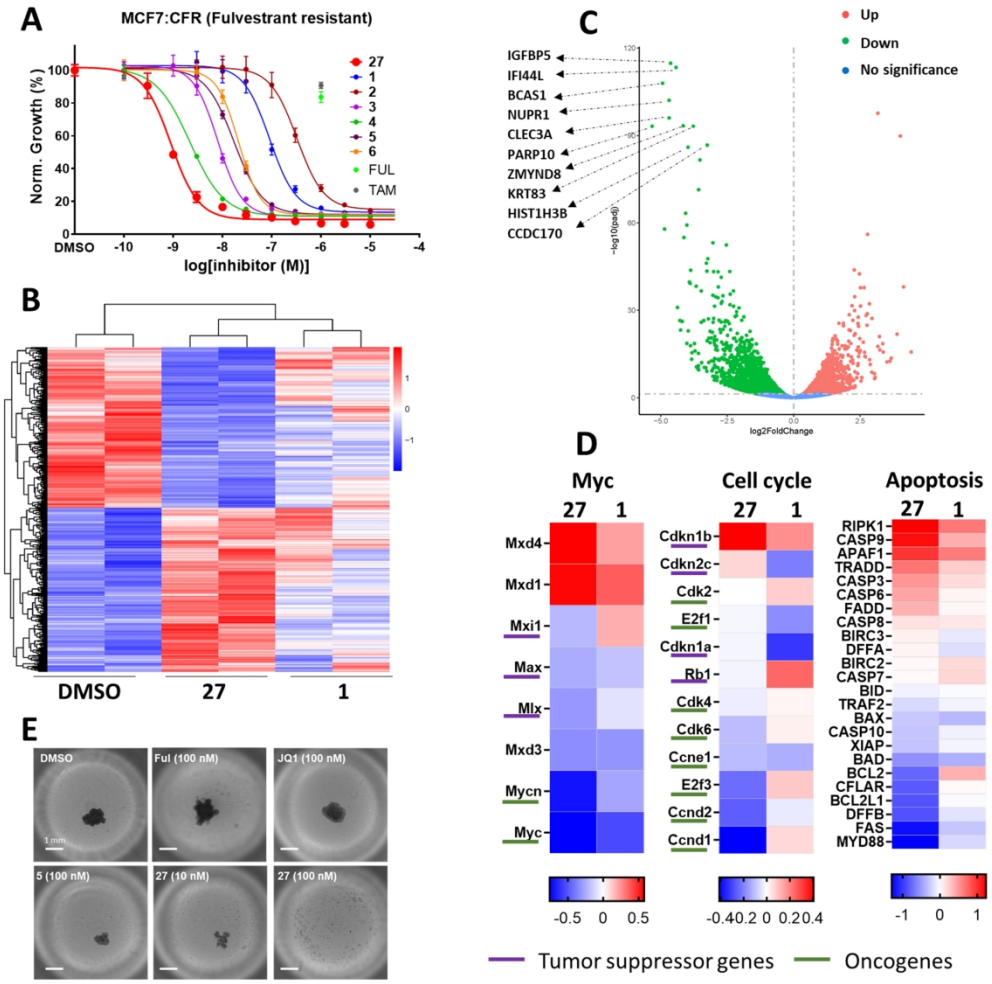

Figure 4. Inhibition of growth and transcriptional regulation of MCF-7:CFR cells by 27 compared to other BET inhibitors. A) Comparison of compound 27 with benchmark BET inhibitors in cell proliferation assay of MCF-7:CFR 2D cell cultures showing resistance to fulvestrant and 4OHT (TAM) (1 μM). B) Heatmap comparing gene expression changes among control (DMSO), 27 (100 nM) and 1 (100 nM) treated MCF-7:CFR cells after 24 h. C) Volcano plot showing top 10 downregulated genes from RNA-seq comparing 27 with vehicle control. D) Log2-fold change of significantly regulated oncogenes and tumor suppressors in various oncogenic (Myc, Cell cycle,) and apoptotic pathways. E) Representative images on day 9 of treatment with DMSO (0.01%), fulvestrant (100 nM), JQ1 (1, 100 nM), AZD5153 (5, 100 nM), and compound 27 (10 and 100 nM) in MCF7:CFR 3D spheroids; scale bar = 1 mm.

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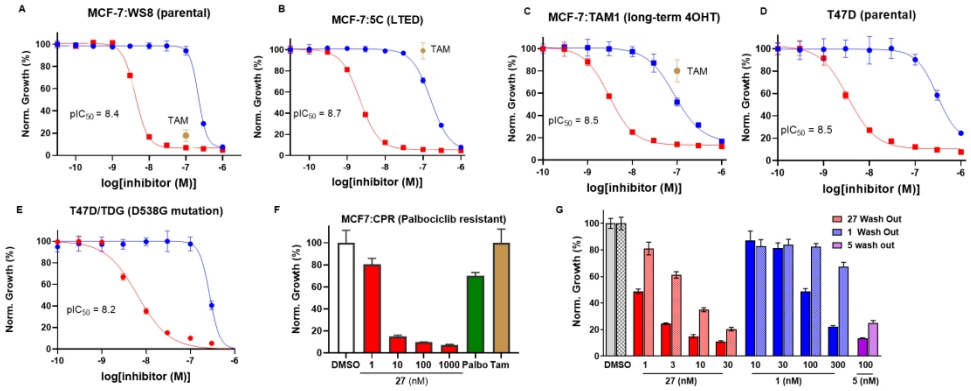

Figure 5. BET inhibitor 27 is a potent antiproliferative agent in breast cancer cell lines resistant to tamoxifen, AIs, CDK4/6 inhibitors, and fulvestrant. (A-E) Concentration-dependent response to compound 27 (red) compared to 1 (blue) and 4OHT (TAM, brown), with potency for 27 annotated. A) 27 dose-dependently inhibited the proliferation of parental MCF-7:WS8 cells that are sensitive to growth inhibition by tamoxifen. Compound 27 dose-dependently inhibited the proliferation of tamoxifen resistant MCF-7:5C (B) and MCF-7:TAM1 (C) cells with superior potency to 1. Compounds 27 and 1 dose-dependently inhibited the proliferation of tamoxifen sensitive parental T47D cells (D) and ESR1 D538G mutant T47D/TDG cells (E), with no significant right-shift in potency in the ESR1 mutant cell line. F) Compound 27 dose-dependently inhibited the proliferation of MCF-7:CPR cells that are resistant to tamoxifen (brown 1 μM) and palbociclib (green 1 μM). G) Comparison of 27 with 1 and 5 in fulvestrant-resistant MCF-7:CFR cell proliferation assay using drug washout after 3 h, by replacing media after incubation with drug for 3 h and measuring cell viability after 5 days; solid bars show no washout; hashed bars show washout treatment.

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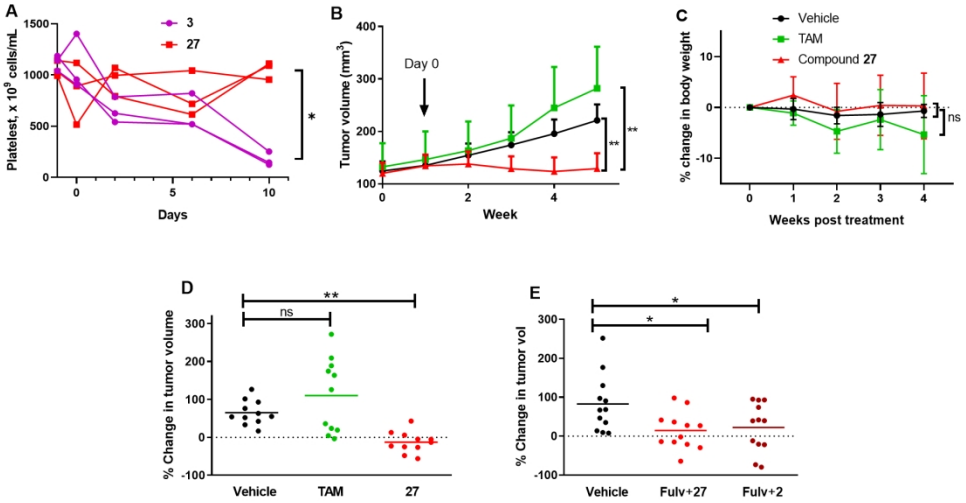

Figure 6. Platelet count in rats and inhibition of tumor growth in mice of tamoxifen-resistant MCF7:TAM1 cell-derived xenografts. A) Platelets counts from rats administered 27 (25 mg/kg) or 3 (3 mg/kg) starting at Day 0 showing data for individual rats (n=3): $p < 0.05$ at Day 10 by two-tailed paired t-test. B) Tumors grown to 0.32 cm2 were treated with vehicle, TAM (1.5 mg/mouse/day), or compound 27 (HCl salt) (50 mg/kg/day) by daily oral gavage. Data show mean and S.D. C) Average body weight changes compared to pretreatment. D) Changes in individual tumor size at 4 weeks. E) Low dose fulvestrant (1 mg/mouse/week by s.c.) in combination with 2 50 mg/kg/day (formulated in 0.5% (w/v) HPMC/TWEEN80 1% in water69) or 27 30 mg/kg/day (formulated in 10% PEG400/90% HP-β-CD (10%)). Changes of individual tumor size at 4 weeks were plotted: data show mean and S.D. with one-way ANOVA analysis with Dunnett's multiple comparisons test.

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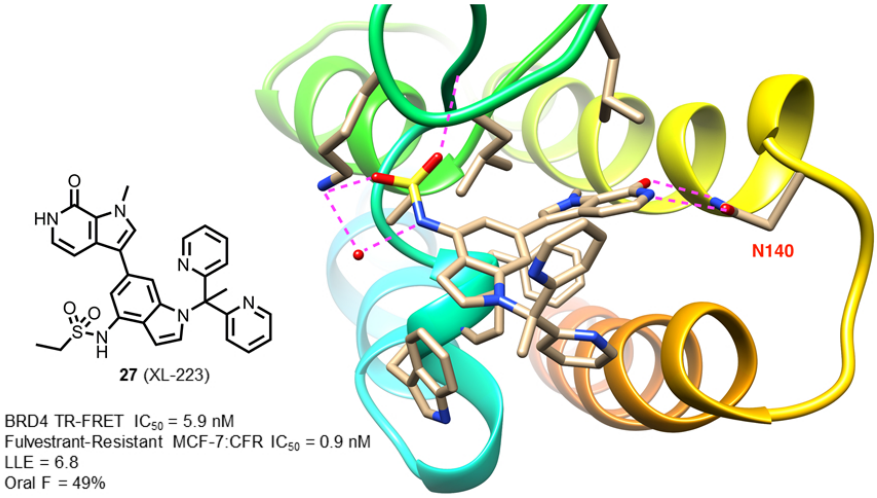

Table of content

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Novel pyrrolopyridone bromodomain and extra-terminal motif (BET) inhibitors effective in endocrine-resistant ER+ breast cancer with acquired resistance to fulvestrant and palbociclib

Yangfeng Li,[‖] Jiong Zhao,[≠] Lauren M. Gutgesell,[≠] Zhengnan Shen,[‖] Kiira Ratia,[‖#] Katherine Dye,[‖] Oleksii Dubrovyskyii,[‖] Huiping Zhao,[≠] Fei Huang,[‖] Debra A. Tonetti,[≠] and Gregory R. J. Thatcher,*[≠‖] Rui Xiong,*[≠‖]

[‖] UICentre (drug discovery @ UIC), University of Illinois at Chicago, 833 S Wood St, Chicago, Illinois 60612

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Research Resources Center, University of Illinois at Chicago, Chicago, Illinois 60612

Keywords: Breast cancer, fulvestrant resistance, estrogen receptor, palbociclib resistance, BET inhibitor

Abstract

Acquired resistance to fulvestrant and palbociclib is a new challenge to treatment of estrogen receptor positive (ER+) breast cancer. ER is expressed in most resistance settings, thus, bromodomain and extra-terminal protein inhibitors (BETi) that target BET-amplified ER-mediated transcription have therapeutic potential. Novel pyrrolopyridone BETi, leveraged novel interactions with L92/L94 confirmed by a co-crystal structure of **27** with BRD4. Optimization of BETi using growth inhibition in fulvestrant-resistant (MCF-7:CFR) cells was confirmed in endocrine-resistant, palbociclib-resistant, and *ESR1* mutant cell lines. **27** was more potent in MCF-7:CFR cells than six BET inhibitors in clinical trials. Transcriptomic analysis differentiated **27** from the benchmark BETi, JQ-1, showing: downregulation of oncogenes; and upregulation of tumor suppressors and apoptosis. The therapeutic approach was validated by oral administration of **27** in orthotopic xenografts of endocrine-resistant breast cancer in monotherapy and in combination with fulvestrant. Importantly, at an equivalent dose in rats, thrombocytopenia was mitigated.

INTRODUCTION

Breast cancer is the second leading cause of cancer death in women worldwide.[1] One out of every eight women in the U.S. will develop invasive breast cancer during her lifetime. Approximately 70% of all breast cancers are estrogen receptor positive (ER+), where dysregulated ER signaling fuels cancer growth. [2] The mainstay of disease control is endocrine therapy using tamoxifen and aromatase inhibitors (AIs) (e.g. anastrazole, letrazole). Tumors with acquired resistance to tamoxifen and AIs maintain functional ER signaling,[3, 4] and therefore ER remains a therapeutic target. Thus, the selective ER degrader (SERD)[5], fulvestrant, has become a key player in the therapeutic response to endocrine resistant, metastatic breast cancer. The standard-of-care in metastatic breast cancer has changed rapidly with the introduction of CDK4/6 inhibitors, such as palbociclib[6], used in 1st line therapy in combination with fulvestrant or AIs.[7-9] However, the majority of patients with ER+ metastatic breast cancer will experience disease progression.

The emerging clinical resistance to both fulvestrant and CDK4/6 inhibitors, presents a new and immediate challenge for targeted drug discovery. Fulvestrant is effective in the endocrine resistant setting; that is, in ER+ breast cancer resistant to tamoxifen and AIs.[10-14] Therefore, the specific mechanisms of resistance to fulvestrant, though not fully defined, are clearly different from mechanisms that drive resistance to endocrine therapy. The optimization of targeted therapies for fulvestrant-resistant breast cancer must therefore be driven by activity in fulvestrant-resistant ER+ breast cancer cells. This work is the first example of iterative lead optimization driven by anti-proliferative activity in fulvestrant and endocrine-resistant ER+ breast cancer cells. Moreover, activity was confirmed in an endocrine-resistant cell line stably resistant to the CDK4/6 inhibitor palbociclib.

Bromodomain and extra-terminal domain (BET) family proteins are “readers” of epigenetic regulation that recognize acetylated lysines in histones H3 and H4.[15-18] The BET family is

comprised of bromodomain-containing proteins, BRD2, BRD3, BRD4, and bromodomain testis-specific protein (BRDT), all featuring two tandem N-terminal bromodomains and an extra C-terminal domain (ET).[19] Each bromodomain is comprised of ~110 amino acids that structurally consists of 4 α-helices (αZ, αA, αB, and αC) and 2 loops, connecting αZ and αA (ZA loop) and αB and αC (BC loop).[20] Early research on BET proteins, predominantly BRD4, revealed a transcriptional coactivator function in enhancer and super-enhancer complexes,[21] facilitating transcription from master regulatory transcription factors, such as MYC, to RNA polymerase II (RNAPII).[22] BET proteins then trigger RNAPII release from the promoter,[23] driving transcriptional elongation and transcription of oncogenes. BRD4 fusion proteins were recognized to drive the aberrant growth of lung, head, and neck cancer.[24]

Recently, BRD4 was reported to be recruited to distal estrogen response elements (ERE) and to coordinate with ERα in a transcriptional complex that regulates gene expression in breast cancer.[25] The first generation BET inhibitor, JQ1 (**1**; Fig. 1)[26], was reported to suppress ER signaling and to inhibit estrogen-mediated breast cancer cell growth.[27, 28] Specifically, BRD4 recruitment to the transcriptional complex formed on ERα binding, histone acetylation, and cohesin recruitment is necessary for elongation-associated phosphorylation of RNAPII (Fig. 1A). Since ER is expressed and plays a ligand-independent role in supporting growth of endocrine-resistant breast tumors, these findings support the use of a BET inhibitor in endocrine and fulvestrant-resistant breast cancer.

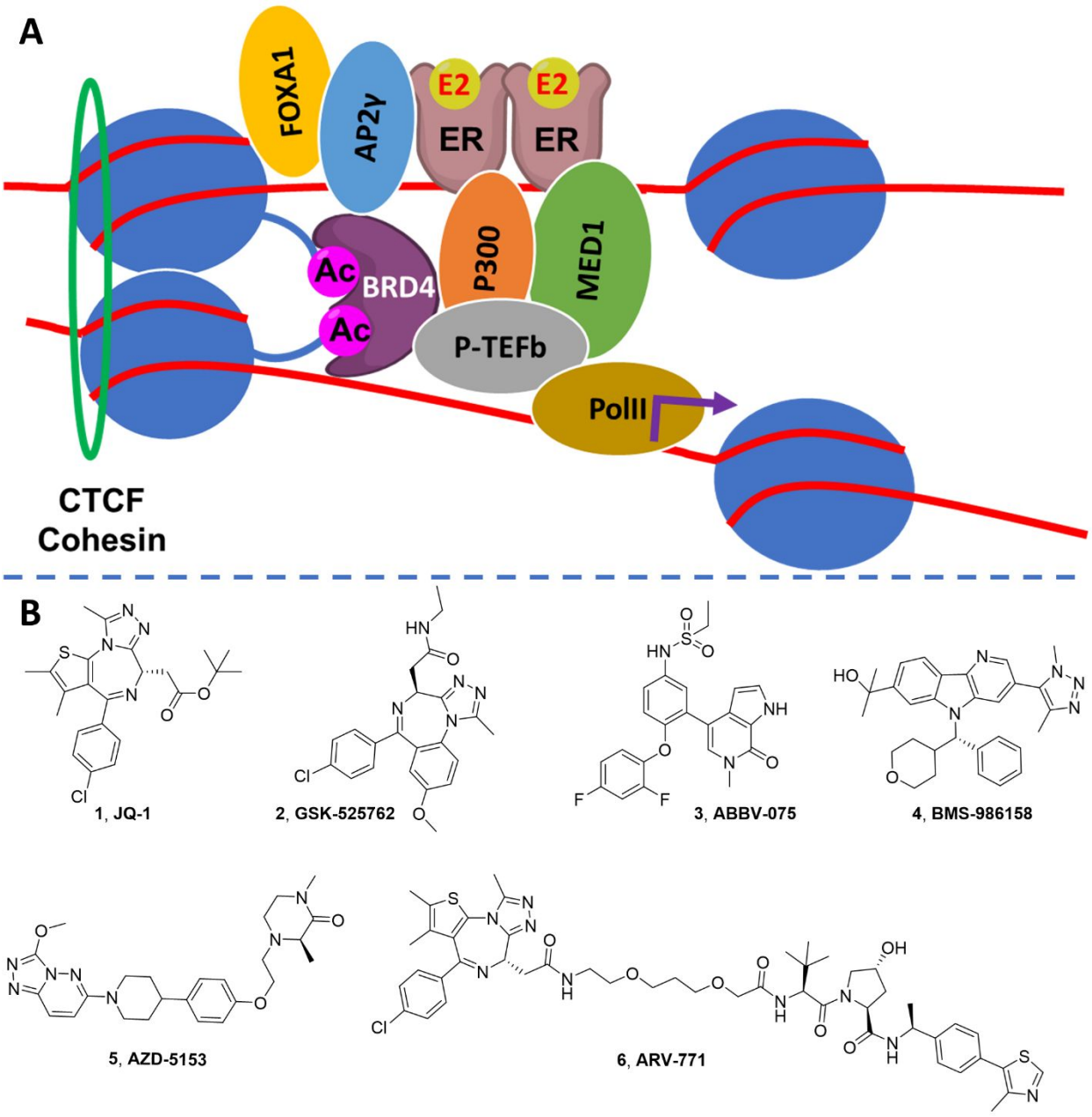

A
FOXA1
AP2γ
E2
ER
E2
ER
Ac
BRD4
Ac
P300
MED1
P-TEFb
PolII
CTCF
Cohesin
B
1, JQ-1
2, GSK-525762
3, ABBV-075
4, BMS-986158
5, AZD-5153
6, ARV-771

Figure 1. BRD4 is a coregulator of ER signaling in breast cancer. A) ER+ breast cancer is a transcriptionally-driven cancer: E2-liganded ER localizes to ERE forming a transcriptional complex, including proteins such as MED1, P300 and FOXA1; BRD4 recruitment to this complex at acetylated histones and cohesin recruitment, causes transcriptional elongation via P-TEFb and RNA polymerase II. Targeting the ER transcriptional complex through BRD4 provides an alternative approach either as monotherapy or in combination with SERDs to treat ER+ breast cancer; B) Structures of benchmark BET inhibitors.

Small molecules with diverse bromodomain-targeting scaffolds[29-43] have been reported over the last decade, since the discovery of **1**,[26 44] including >20 BET inhibitors that entered clinical trials for various therapeutic applications such as advanced solid tumors and hematological malignancies (Fig. 1B).[45] Benzodiazepine-based GSK-525762 (**2**),[31] pyridine-based ABBV-075 (**3**),[32] triazole-based BMS-986158 (**4**),[33] and bivalent AZD-5153 (**5**)[34] all feature an acetylated lysine mimic that forms a hydrogen bond with a conserved Asn140 in the BD1 domain of BRD4. Herein, we report a novel series of potent, pyrrolopyridone BET inhibitors, leveraging a novel interaction in the BD1 domain, confirmed by co-crystal structures.

A unique iterative optimization strategy, driven by antiproliferative potency in a fulvestrant-resistant ER+ cell line (MCF-7:CFR), led to BET inhibitor **27**, which demonstrated superior potency to other clinical-stage BET ihhibitors in this cell line. **27** was also shown to be a potent inhibitor of growth of breast cancer cells resistant to tamoxifen, aromatase inhibitors, and palbociclib. This strategy was designed specifically to target emerging treatment-resistant ER+ breast cancers and was supported by transcriptomic analysis in MCF-7:CFR cells identifying downregulated oncogenes and upregulated tumor suppressor and pro-apoptotic genes. **27** was orally bioavailable and inhibited growth of endocrine-resistant tumors *in vivo,* without thrombocytopenia.

DESIGN & OPTIMIZATION

Initial structure-based design of inhibitors was driven by literature crystal structures. The design strategy was validated by obtaining a new crystal structure of our best BET inhibitor bound

to BRD4. Inhibitor optimization was driven by two assays: 1) growth inhibition of the MCF-7:CFR cell line; and 2) measurement of binding to BRD4-BD1 using TR-FRET. The ER+ MCF-7:CFR cell line is resistant to endocrine therapy and resistant to treatment with fulvestrant, modeling the clinical resistance setting wherein ERα continues to drive tumor survival, but has acquired resistance to all ER-targeted clinical therapeutics. The MCF-7:CFR cell line was obtained by long-term exposure of the endocrine-resistant MCF-7:5C cell line to fulvestrant and provides a stable phenotype across multiple passages. This is a unique optimization strategy directed at addressing the emerging clinical resistance in metastatic breast cancer.

A pyrrolopyridone scaffold was identified from virtual screening by Vidler et al.[46, 47] The pyrrolopyridone ring acts as an acetylated lysine mimic (Fig. 2A), forming a bidentate hydrogen-bond with Asn140 (BRD4-BD1). We postulated that the 3 position, amenable to functionalization, would permit optimization and drew inspiration from earlier studies: the sulfonamide group of **3**[32] and bromosporine (**8**);[48, 49] and, the heteroaryl moiety of I-BET151 (**9**)[50]. The sulfonamide group of **8** is proposed to interact with the ZA loop, whereas the heteroaryl moiety of **9** is proposed to bind to the hydrophobic WPF shelf; both important structural features of the BD1 binding site (Fig. 2B). Compound **10**, combining these two key design elements in a “trans” arrangement, was anticipated to engage both the ZA loop and WPF shelf, increasing affinity and potency (Fig. 2A,B). Superposition of the predicted docking pose of **10** with that of **9** from a co-crystal structure with BRD4-BD1 (pdb: 3ZYU) showed that **10** captures most of the binding interactions of **9** with BRD4-BD1; moreover, **10** also forms an extra hydrogen bond with Asn140 that might enhance the binding affinity (Fig. 2C). Further designs were also tested in this preliminary round of SAR, including the regioisomers **12-14,** benzimidazoles **15, 16,** and indazoles **17, 18.**

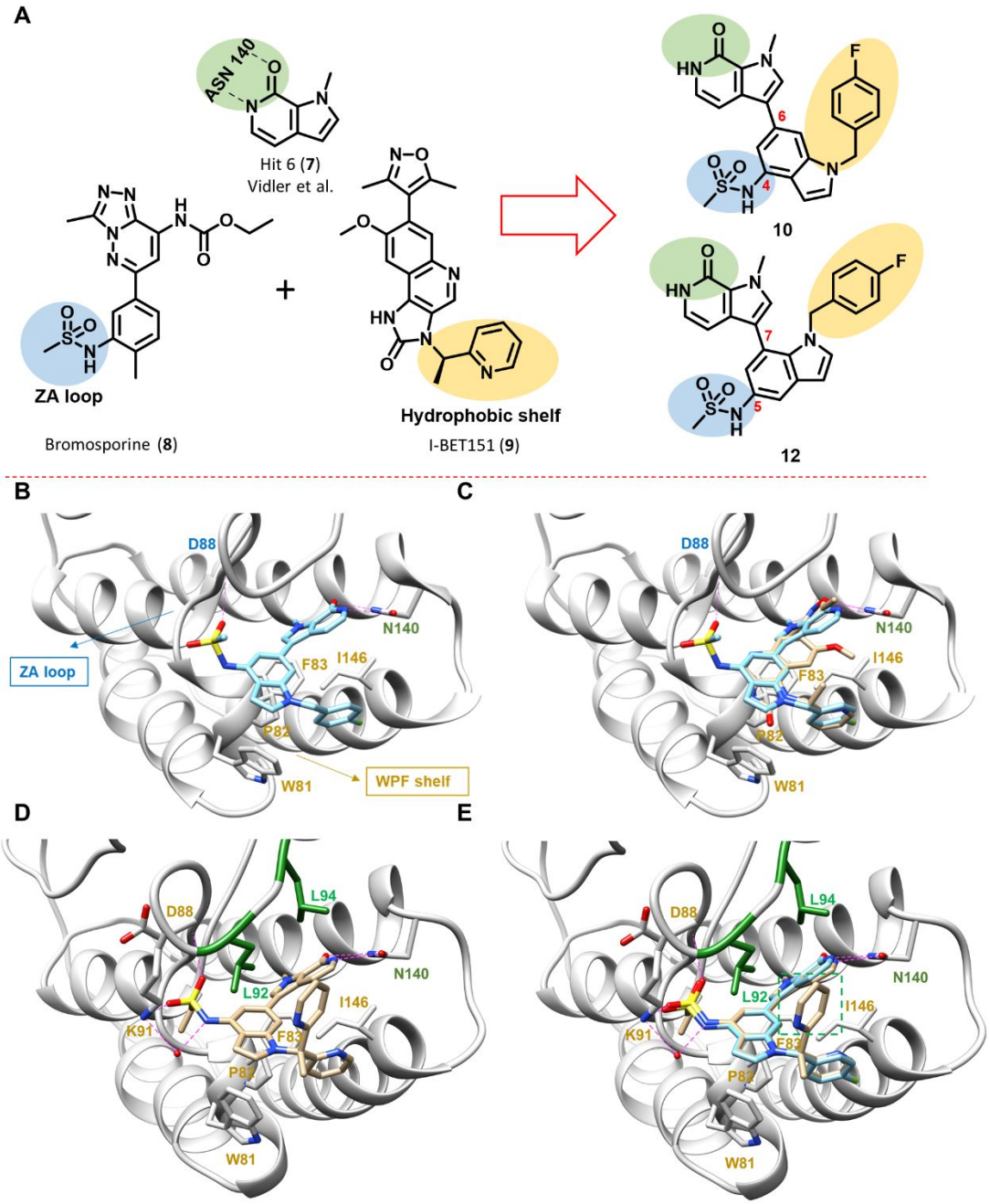
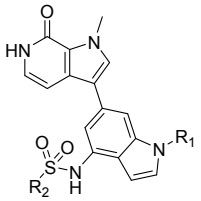


Figure 2. Design of pyrrolopyridone-based BET inhibitors. A) Design concept inspired by earlier reported BET inhibitors: 1) pyrrolopyridone, **7**, identified from a virtual screen forms

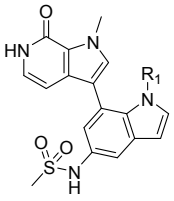
bidentate hydrogen-bonding interaction with Asn140, highlighted in green, 2) sulfonamide group of **8** interacts with ZA loop via a putative hydrogen-bonding interaction, highlighted in blue; and 3) the heteroaryl moiety of **9** interacts with the hydrophobic WPF shelf, highlighted in yellow. Combining these design parameters leads to the design of **10** and **12** as novel BET inhibitors; B) Predicted binding pose of compound **10** binding to BRD4-BD1 using Glide (pdb: 3ZYU): key interacting residues from docking are highlighted and labeled; C) Overlay of the predicted binding pose of **10** (blue) with **9** (beige) from pdb: 3ZYU. D) New co-crystal of compound **27** bound to BRD4-BD1 (pdb: 6P05), showing four hydrogen bonds with N140, D88 and K91, and one additional water-mediated hydrogen bond with K91. The "leucine clamp" (L92 and L94) is highlighted in green and other key interacting residues are highlighted and labeled in red. E) Overlay of the predicted binding pose of **10** (blue) with the experimentally confirmed binding pose of **27** (beige) clearly showing the second pyridine ring of **27** (boxed in green) that forms a hydrophobic interaction with L92 and L94.

The design strategy for compound **10** was substantiated by TR-FRET assay of binding to BRD4-BD1 (IC_{50} = 96 nM), while the positive control **1**, showed slightly weaker binding affinity in this assay (IC_{50} = 199 nM) (Table 1). In contrast, the 7-pyrrolopyridone substituted indole **12**, the regioisomer of **10**, showed a dramatic loss of potency (IC_{50} = 1.18 μM). In MCF-7:CFR cells, the high potency of **10** in inhibiting cell growth (IC_{50} = 17 nM) was confirmed, comparing favorably with **1** (IC_{50} = 90 nM), and in accord with TR-FRET binding data. A further attempt to optimize the phenyl ring in the 7-pyrrolopyridone substituted indole series yielded compounds **13** and **14**, which again lacked potency in both binding and growth inhibition assays (Table 1). Two further, alternative heterocycle designs were studied: the benzimidazole **15** exhibited a significant decrease in BRD4-BD1 binding by TR-FRET and reduced potency in MCF-7:CFR cells (IC_{50} = 0.7 μM). The benzimidazole **16** suffered a further loss of potency indicating a preference for a mono-sulfonamide in interacting with the ZA loop. Finally, **17** and **18** were designed to probe the effect of an indazole scaffold in comparison to the indoles, **10** and **11**, showing similar potency in biochemical and cell-based assays.

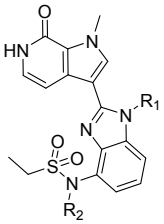
Table 1. Biochemical and Cellular Potency of Compounds **10-18**



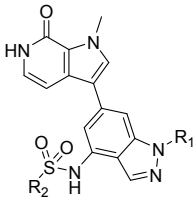
10, **11**



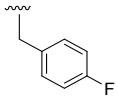
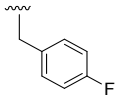
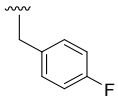
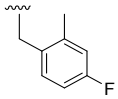
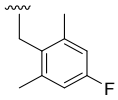
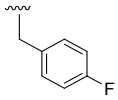
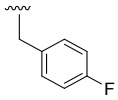
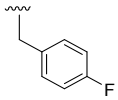
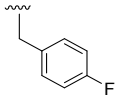
12 - **14**



15, **16**



17, **18**

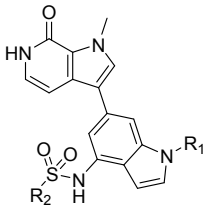
Compound	R_1	R_2	TR-FRET BD1 IC_{50} (nM)[a]	MCF-7:CFR IC_{50} (nM)[b]	LLE[c]
1	-	-	199 ± 25.6	90 ± 4.4	-
10		Me	96 ± 11.8	17 ± 0.8	4.677
11		Et	48 ± 9.0	56 ± 1.6	3.630
12		-	1181 ± 419	1994 ± 96	2.608
13		-	1000 ± 326	3577 ± 284	1.905
14		-	1400 ± 715	526 ± 20	2.289
15		H	1434 ± 749	708 ± 27	3.018
16		SO_2Et	NA	1780 ± 87	2.379
17		Me	89 ± 9.9	68 ± 3.0	4.520
18		Et	54 ± 5.4	25 ± 0.9	4.426

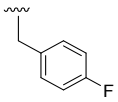
[a]TR-FRET assay with BRD4(1) bromodomain IC_{50} values are reported as mean ± SD. [b]Cell survival was normalized to DMSO control (100 %) and JQ1 (3 μM) as 0% assessed in a 5-day cell proliferation assay with one time drug treatment. Data shows mean ± SEM from at least two cell passages (triplicates in each passage). [c]LLE (LipE) = pIC_{50} – cLogP, IC_{50} of growth inhibition was used for calculations.

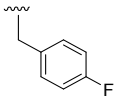
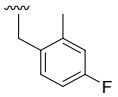
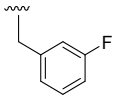
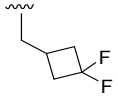
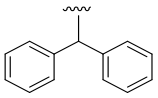
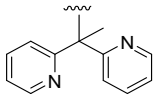
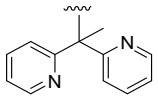
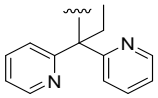
Ligand-lipophilicity efficiency (LLE)[51, 52] for compounds in the first round of SAR was calculated using cell based potency to estimate drug-likeness (Table 1), with **10, 17,** and **18** showing superior metrics. However, a preliminary pharmacokinetic study showed that **10** had oral bioavailability clearly superior to the indazoles (data not shown); therefore, 6-pyrrolopyridone substituted indole **10** was selected for further optimization.

Compounds **19** - **22** were designed to alter the torsional angle between the phenyl ring and the indole moiety by derivatizing the phenyl ring, leading to comparable potency to **10** (Table 2), suggesting that the WPF region tolerates varied ring substituents. Efforts to improve potency and increase sp^3 count were made by replacement with nonaromatic substituents, namely oxetane or difluorocyclobutane rings, as in compounds **23** and **24**, respectively. These changes were tolerated without significant loss of potency and with an improvement in the LLE metric. To overcome the potential metabolic liability of the benzylic position of **10**, we introduced a second phenyl group at this position in compound **25**, which gratifyingly led to improved potency relative to **10**. The high potency in inhibition of MCF7:CFR cell growth (IC_{50} = 15 nM) was counterbalanced by a less favorable LLE.

Table 2. Biochemical and Cellular Potency of substituted indoles



Compound	R_1	R_2	TR-FRET BD1 IC_{50} (nM) [a]	CFR IC_{50} (nM) [b]	LLE[c]
1	-	-	199 ± 25.6	90 ± 4.4	-
10		Me	96 ± 11.8	17 ± 0.8	4.677

11		Et	48 ± 9.0	56 ± 1.6	3.630
19		Et	158 ± 24.4	35 ± 1.1	3.079
20		Me	175 ± 23.5	49 ± 1.9	3.462
21		Me	600 ± 190.0	119 ± 4.7	3.383
22		Me	174 ± 21.9	70 ± 2.6	4.062
23		Me	180 ± 27.9	134 ± 6.4	5.789
24		Me	161 ± 29.6	62 ± 2.3	4.858
25		Et	83 ± 10.9	15 ± 0.5	2.997
26		Me	12.9 ± 0.7	10 ± 0.7	6.297
27		Et	5.9 ± 0.3	0.9 ± 0.03	6.814
28		Et	9.4 ± 0.4	2.2 ± 0.09	5.897

[a] TR-FRET assay with BRD4(1) bromodomain IC_{50} values are reported as mean ± SD. [b]Cell survival was normalized to DMSO control (100 %) and JQ1 (3 μM) as 0% assessed in a 5-day cell proliferation assay with one time drug treatment. Data shows mean ± SEM from at least two cell passages (triplicates in each passage). [c]LLE (LipE) = pIC_{50} – cLogP, IC_{50} of growth inhibition was used for calculations.

The gain in potency of **25** suggested the existence of extra stabilizing interactions with the second phenyl ring, which could be further optimized. The less favorable LLE also indicated the need to explore heterocycles in place of benzene rings. A significant improvement in biochemical and cell-based assay potency was achieved by replacing the di-phenyl substituent with a bipyridyl

analog and simultaneously converting the benzylic position into a quaternary carbon center in compounds **26**-**28**. The ethyl substitution at this position in compound **28** was inspired by molecular modeling indicating a small hydrophobic pocket; however, compound **28** was inferior to **26** and **27** in binding potency and LLE, suggesting a binding penalty in further modifying this site. An exemplar TR-FRET binding curve is shown in Fig. 3A for compound **27**.

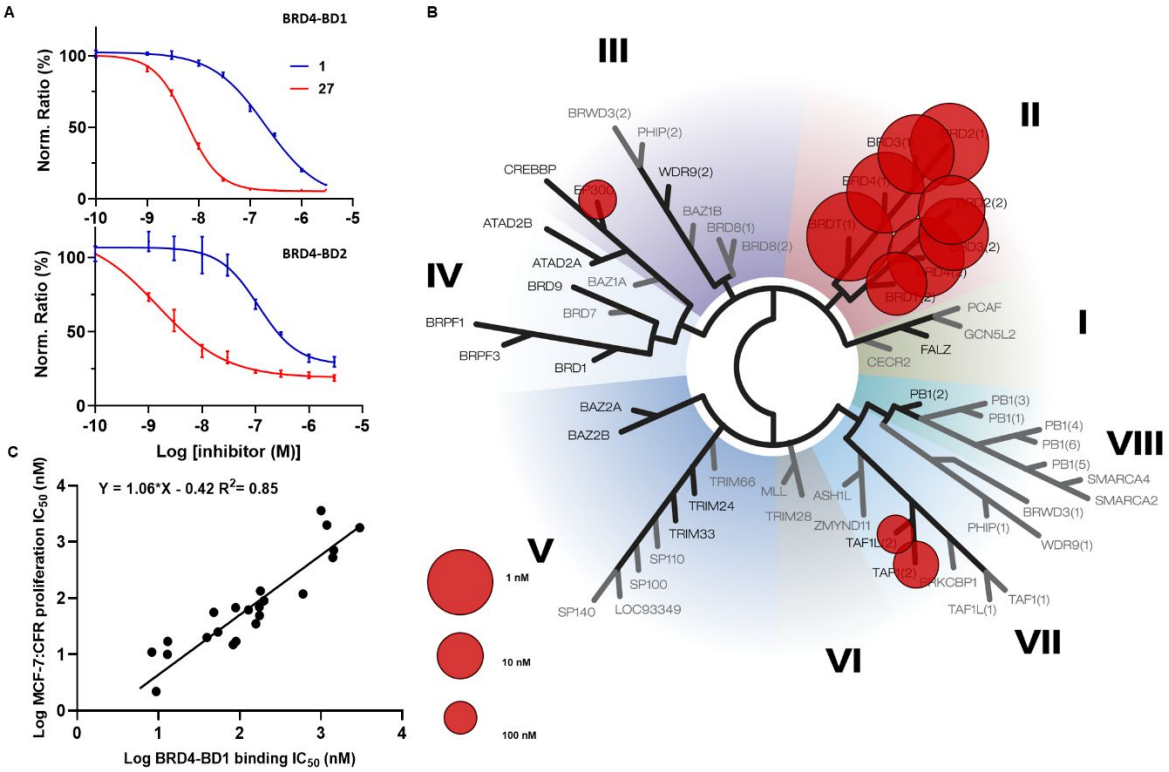


Figure 3. **Binding to BRD4-BD1 correlates with cellular antiproliferative potency.** A) BRD4-BD1 and BRD4-BD2 binding measured by TR-FRET for **1** compared to **27** (data normalized to DMSO as 100%); B) BromoScan of compound **27** against 40 bromodomain proteins; C) Correlation between BRD4-BD1 binding (TR-FRET) assay and potency for growth inhibition of MCF-7:CFR cells.

With compound **27** demonstrating a good combination of potency and LLE, we obtained a co-crystal structure of **27** in complex with BRD4-BD1 to better understand the structural basis of binding to BRD4 and the high potency in biochemical and cell growth inhibition assays (Fig. 2D). The structure confirmed our initial design hypotheses: 1) the pyrrolopyridone scaffold forms a bidentate hydrogen-bonding interaction with Asn140; 2) the sulfonamide moiety inserts into the

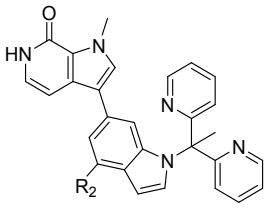
ZA loop and its neighboring residues; and 3) one pyridine ring interacts with the hydrophobic WPF shelf. However, less anticipated was the observed hydrogen-bonding network induced by the sulfonamide moiety both directly with Asp88 and Lys91, and via an additional water-mediated hydrogen bond with Lys91.

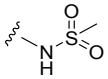
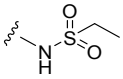
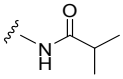
In addition to the sulfonamide-induced hydrogen bonding network, we also observed that the second pyridine ring forms an edge-to-face interaction with a “leucine clamp” (Leu92 and Leu94) in the ZA loop (Fig. 2E), which we postulated to contribute to the high affinity and potency of compound **27**. A similar interaction was previously reported (Supplemental Fig. S1).[53] In compound **27,** the sterically congested, and conformationally restricted methinyl pyridine likely contributes to the increased binding affinity. In summary, our new co-crystal structure showed a unique edge-to-face interaction with the “leucine clamp” that boosted the potency by an order of magnitude. This interaction could be incorporated into other scaffolds, serving as a generalizable design strategy towards next-generation BET inhibitors.

The binding interactions of **27** at the BD1 site revealed a hydrogen-bond network with the ZA loop, involving Asp88 and Lys91. Interestingly, Lys91 is one of the key residues that differentiate the BD1 domain from the BD2 domain of BET proteins.[54] Therefore, these interactions provide an opportunity for further optimization for both potency and selectivity. We explored substitutions at the 4-position of the indole ring in the bipyridyl series. Replacing the sulfonamide with amides in **29** and **30** led to BET inhibitors with similar potency, but an inferior LLE, whereas replacement with a 2-hydroxypropyl group resulted in 2- to 3-fold loss of potency in biochemical and cell-based assays (Table 3). Most compounds, except **31**, showed comparable binding between BD1 and BD2 in TR-FRET assays, indicating a similar hydrogen bonding interaction with the ZA loop. Compound **31** displayed a sixfold selectivity for BD1 versus BD2, likely resulting from a selective interaction with Lys91 and provided the foundation for BD1-

selective inhibitors that will be reported in a separate paper. However, given the excellent potency and superior LLE, compound **27** was selected for further study.

Table 3. Exploration of 4-position of indole-based analogs



Compound	R_2	TR-FRET BD1 IC_{50} (nM) [a]	TR-FRET BD2 IC_{50} (nM) [a]	CFR IC_{50} (nM) [a]	LLE[b]
1	-	199 ± 25.6	126 ± 18.8	90 ± 4.4	-
26		12.9 ± 0.7	12.7 ± 1.2	10 ± 0.7	6.297
27		5.9 ± 0.3	1.7 ± 0.4	0.9 ± 0.03	6.814
29		13 ± 1.0	62.8 ± 20.5	17 ± 0.8	4.968
30		8.3 ± 0.5	21.6 ± 4.1	11 ± 0.5	5.995
31		39.6 ± 4.2	220 ± 82.2	20 ± 0.8	5.297

[a] TR-FRET assay with BRD4(1) bromodomain IC_{50} values are reported as mean ± SD. [b]Cell survival was normalized to DMSO control (100 %) and JQ1 (3 μM) as 0% assessed in a 5-day cell proliferation assay with one-time drug treatment. Data shows mean ± SEM from at least two cell passages (triplicates in each passage). [c]LLE (LipE) = pIC_{50} – cLogP, IC_{50} of growth inhibition was used for calculations.

MECHANISM AND EFFICACY IN FULVESTRANT RESISTANCE

The ER+ MCF-7:CFR cell line, resistant to tamoxifen, fulvestrant, and AIs, drove the iterative optimization of BET inhibitors leading to **27**. To test the dependence of growth inhibition of MCF-7:CFR cells on binding to BRD4 over other bromodomain proteins, the binding affinity for BRD4-BD1 from TR-FRET was plotted against antiproliferative potency in MCF-7:CFR cells (Fig.

3C). The excellent correlation (R^2 = 0.85), observed across the 23 BET inhibitors created and studied herein, strongly supports the hypothesis that growth inhibition in MCF-7:CFR cells is driven by on-target engagement of BD1. In an orthogonal BROMOscan binding assay against a panel of 40 bromodomains, **27** showed high potency and selectivity (K_d < 1 nM) across BET family proteins (Fig. 3B & Table S1). Selectivity was 90-fold for BRD4 binding (full length, short-iso, K_d = 0.1 nM) over TAF1 (K_d = 9.1 nM) and 370-fold for BRD4 binding over EP300 (K_d = 37 nM). Selectivity for BRD proteins over all other bromodomain proteins, was >400 fold. There is no compelling evidence for EP300 inhibition contributing to the antiproliferative actions of **27** in MCF-7:CFR cells; however, it is noted that NEO2734, a BET inhibitor with activity at EP300, was recently reported to have efficacy in a prostate cancer xenograft model.[55]

Cell cultures grown as 3D spheroids better reflect the influence of a tumor microenvironment in supporting cell survival; therefore, it was important to test our observations in 2D cell cultures by studying effects in MCF-7:CFR cells in 3D. Cells were grown as spheroids in low adhesion plates for 1 day, before addition of compound or vehicle, and cell viability was measured at day 9 (Fig. 4A). Neither 4-hydroxytamoxifen (4OHT), nor fulvestrant inhibited MCF-7:CFR cell growth, as anticipated for a tamoxifen-resistant cell line passaged for a year in media containing fulvestrant. BET inhibitors **1** and **5** significantly inhibited growth and at the same concentration (100 nM), **27** caused complete loss of spheroid viability, compatible with the observed high potency in 2D cultures.

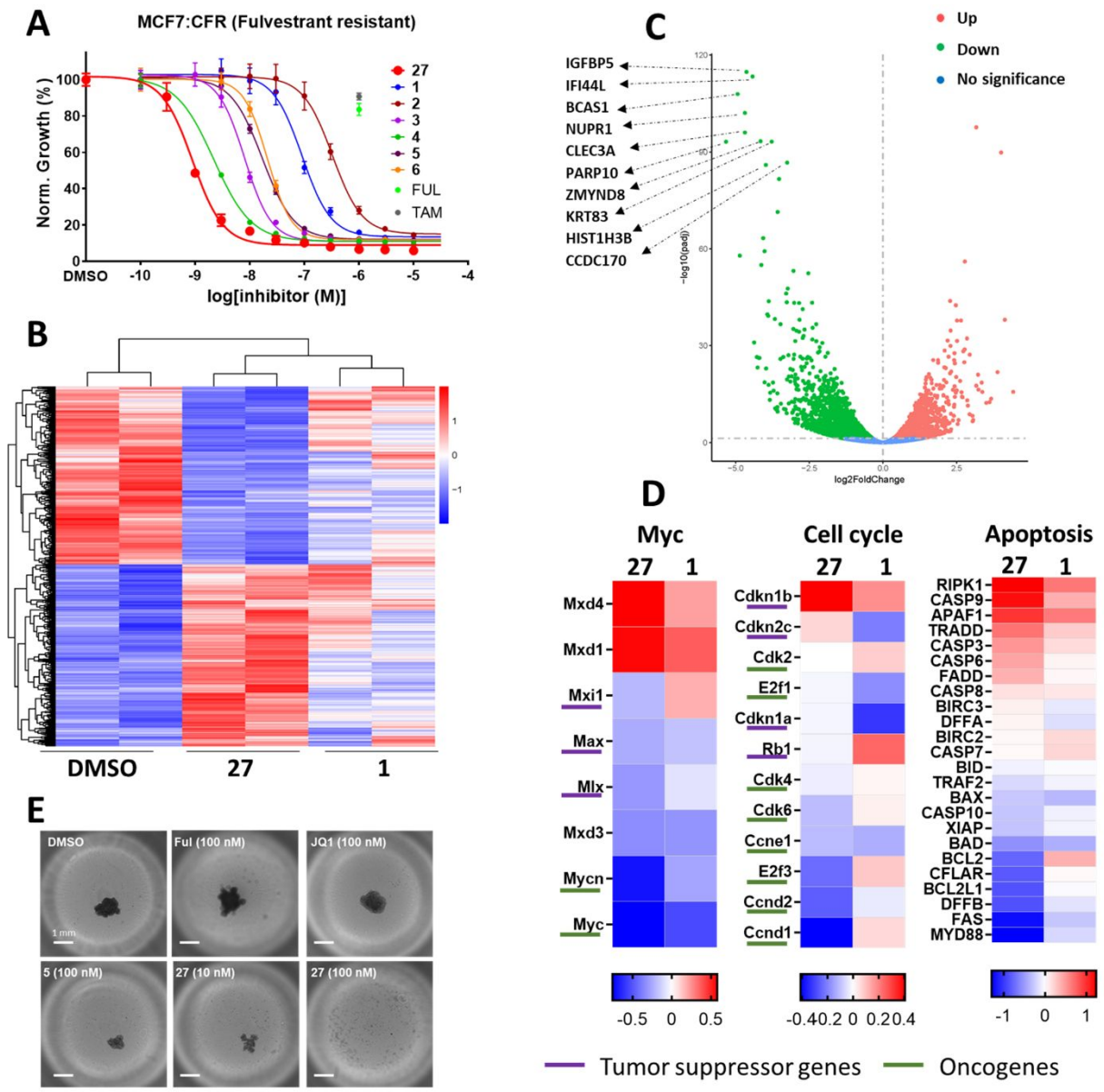


Figure 4. Inhibition of growth and transcriptional regulation of MCF-7:CFR cells by 27 compared to other BET inhibitors. A) Comparison of compound **27** with benchmark BET inhibitors in cell proliferation assay of MCF-7:CFR 2D cell cultures showing resistance to fulvestrant and 4OHT (TAM) (1 μM). B) Heatmap comparing gene expression changes among control (DMSO), **27** (100 nM) and **1** (100 nM) treated MCF-7:CFR cells after 24 h. C) Volcano plot showing top 10 downregulated genes from RNA-seq comparing **27** with vehicle control. D) Log2-fold change of significantly regulated oncogenes and tumor suppressors in various oncogenic (Myc, Cell cycle,) and apoptotic pathways. E) Representative images on day 9 of treatment with DMSO (0.01%), fulvestrant (100 nM), JQ1 (**1**, 100 nM), AZD5153 (**5**, 100 nM), and compound **27** (10 and 100 nM) in MCF7:CFR 3D spheroids; scale bar = 1 mm.

In the MCF-7:CFR cell line, the potency of compound **27** was compared to: JQ1 (**1**); four BET inhibitors currently in clinical trials (**2-5**); and the BET-PROTAC **6** (Fig. 4B & Table S3). Compound **27** was more potent than all other BET inhibitors and degraders tested. The measured potency for growth inhibition by **27** (IC_{50} = 0.9 nM) is compatible with the potency and affinity shown in the Bromoscan and TR-FRET assays. In addition, the observed potency is 10- to 40-fold greater than measured for inhibition of TAF1 and EP300 from Bromoscan data, and therefore is incompatible with growth inhibition driven by inhibition of TAF1 or EP300. The exceptional potency and superiority in growth inhibition of MCF-7:CFR cells by **27** compared to six other BET inhibitors that have reached clinical trials can be explained by the additional interactions of **27** with Lys91 and the leucine clamp. Further, it should also be noted that the MCF-7:CFR cell line was used in the iterative optimization of compound **27**.

To further explore the mechanism of action of **27** in inhibiting growth of fulvestrant-resistant MCF-7:CFR cells, we investigated genome-wide gene expression through RNA-seq, comparing **27** to the benchmark inhibitor **1** (Fig. 4C-E). The genome-wide effect of **27** was significantly greater than **1,** despite both compounds being studied at a concentration > IC_{50} (100 nM). BET inhibitor **27** broadly regulated the gene expression profile in MCF-7:CFR cells, with 1637 genes showing >2-fold downregulation and 897 genes >2-fold upregulation (Fig. 4C). The top 10 genes regulated by **27** were IGFBP5, IFI44L, BCAS1, NUPR1, CLEC3A, PARP10, ZMYND8, KRT83, HIST1H3B and CCDC170 (Fig. 4D). However, we focused our analysis on three gene pathway clusters: Myc signaling; cell cycle regulation; and apoptosis (Fig. 4E).

The oncogenic Myc pathway has been shown to be a critical target of BET inhibitor action in the anti-cancer arena. Consistent with these findings, we found **27** significantly decreased expression of Myc pathway oncogenes such as c-MYC and n-MYC. Focusing on cell cycle genes, **27** increased expression of CDKN1B, a tumor suppressor that prevents the activation of the cyclin-D/CDK4 complex, and **27** also reduced expression of cyclin-D (CCND1) itself. Although

Myc pathway genes were regulated in the same direction by **27** and **1**, cell cycle genes were differentially regulated. Differential regulation was also observed in the RTK/RAS pathway, with NRAS, ERBB3 and c-MET downregulated by **27.** The most significantly downregulated oncogene by **27** is c-MET, the overexpression of which is reported to correlate with poor prognosis and treatment resistance from many reports.[56, 57] BET inhibition has been shown to induce apoptosis in various cancer cells and we observed that **27** upregulated pro-apoptotic genes such as RIPK1, APAF1 and CASP3/6/9 and at the same time decreased expression of anti-apoptotic genes, including BCL2L1 and BCL2. In regulation of estrogen-related genes (Fig. S4), *ESR1* itself was transcriptionally downregulated by **27**.

Together, the RNA-seq analysis of **27** in fulvestrant-resistant breast cancer cells revealed: suppression of Myc pathway signaling; inhibition of cylin-D/CDK4 complex; and a profound shift in the pro- versus anti-apoptotic signaling bias towards apoptosis. Moreover, **27** was clearly differentiated from **1** in its gene regulation profile, despite the concentration of **1** tested being tenfold greater than the IC_{50} for growth inhibition by **1** in MCF-7:CFR cells. This differentiation was amplified in the response of genes controlling cell cycle and apoptosis. These observations are compatible with the superior potency of **27** in inhibiting growth of MCF-7:CFR cells, relative to **1**, and highlight the efficacy of **27** in downregulation of oncogenes and anti-apoptotic genes, and upregulation of tumor suppressor and pro-apoptotic genes.

EFFICACY IN ENDOCRINE AND PALBOCICLIB RESISTANCE

BET inhibitor **27** was further profiled in ER+ cell lines modeling tumors resistant to multiple therapeutic agents in the clinical setting, including resistance to tamoxifen, AIs, and palbociclib. A scheme showing the derivatization of cell lines is provided in Fig. S1. In general, long-term estrogen deprivation (LTED) and long-term exposure to 4OHT (the active metabolite of tamoxifen)

are used to model AI and tamoxifen resistance, respectively. The majority of cell lines obtained by these methods remain ER+ and either lose sensitivity to 4OHT or grow more rapidly in the presence of 4OHT. The parental MCF-7:WS8 cell line, from which all MCF-7 resistant lines were obtained, is sensitive to growth inhibition induced by treatment with 4OHT (Fig. 5A). Compound **27** dose-dependently inhibited the proliferation of these tamoxifen-sensitive cells with potency almost two orders of magnitude superior to benchmark inhibitor **1**. A similar pattern was observed for growth inhibition of the tamoxifen-resistant cell lines, MCF-7:5C and MCF-7:TAM1, wherein the potency of **27** was 1-2 orders of magnitude superior to **1** (Fig. 5B,C).

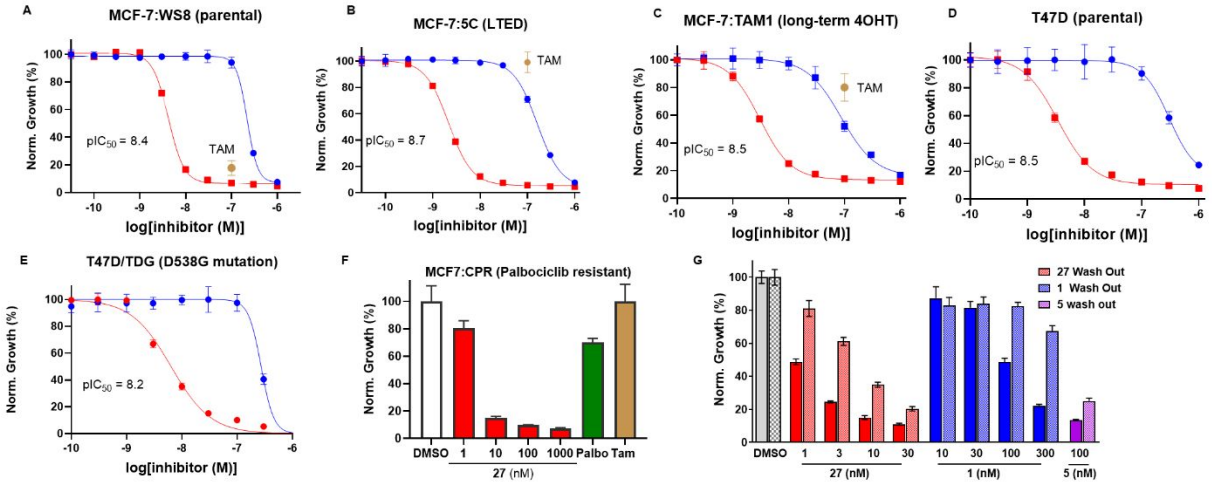


Figure 5. BET inhibitor 27 is a potent antiproliferative agent in breast cancer cell lines resistant to tamoxifen, AIs, CDK4/6 inhibitors, and fulvestrant. (A-E) Concentration-dependent response to compound **27** (red) compared to **1** (blue) and 4OHT (TAM, brown), with potency for **27** annotated. A) **27** dose-dependently inhibited the proliferation of parental MCF-7:WS8 cells that are sensitive to growth inhibition by tamoxifen. Compound **27** dose-dependently inhibited the proliferation of tamoxifen resistant MCF-7:5C (B) and MCF-7:TAM1 (C) cells with superior potency to **1**. Compounds **27** and **1** dose-dependently inhibited the proliferation of tamoxifen sensitive parental T47D cells (D) and *ESR1* D538G mutant T47D/TDG cells (E), with no significant right-shift in potency in the *ESR1* mutant cell line. F) Compound **27** dose-dependently inhibited the proliferation of MCF-7:CPR cells that are resistant to tamoxifen (brown 1 μM) and palbociclib (green 1 μM). G) Comparison of **27** with **1** and **5** in fulvestrant-resistant MCF-7:CFR cell proliferation assay using drug washout after 3 h, by replacing media after incubation with drug for 3 h and measuring cell viability after 5 days; solid bars show no washout; hashed bars show washout treatment.

The tamoxifen-sensitive and endocrine-dependent T-47D cell line was sensitive to growth inhibition by **27** with potency almost 100-fold greater than **1**, demonstrating that the potency of **1** extended beyond MCF-7 breast cancer cell lines (Fig. 5D). The *ESR1* D538G mutation is one of three mutations most commonly observed in patients with acquired resistance to AI therapy, conferring estrogen-independent cancer cell survival.[58] The T47D/TDG cell line expresses only mutant D538G ERα and the parental T47D cell line studied herein provides the appropriate estrogen-dependent comparator (Fig. 5E versus 5D). Compound **27** dose-dependently inhibited the proliferation of T47D/TDG cells with no loss of potency compared to the parent T47D cell line and was again much more potent than benchmark **1.**

Resistance to CDK4/6 inhibitors is emerging in the clinic.[59] Palbociclib is the first and most prescribed drug in this class; therefore, MCF-7:5C cells were passaged in the presence of increasing concentrations of palbociclib for 6 months to obtain an endocrine-independent cell line stably resistant to tamoxifen and palbociclib. In this MCF-7:CPR cell line, potent, concentration-dependent growth inhibition was observed after **27** treatment (Fig. 5F).

BET inhibitor **27** showed both high efficacy and potency (IC_{50} < 10 nM) across all cell lines with significantly superior potency compared to **1** in all cell lines studied and superior potency to a variety of BET inhibitors in the MCF-7:CFR cell line. Although many more potent BET inhibitors have been discovered since the discovery of JQ1 (**1**), this BET inhibitor remains the benchmark with >140 papers annually reporting data on JQ1 from 2017-19.

ADMET & IN VIVO EFFICACY

Profiling of **27** in ADMET and physicochemical studies, demonstrated acceptable solubility (73.4 μM, pH = 7.4) and human hepatocyte stability ($T_{½}$ = 202 mins; clearance rate = 6.86 μL/min/10^6 cells) (Table 4). The IC_{50} for hERG inhibition was 10,000-fold right-shifted from the potency observed for inhibition of breast cancer cell growth, indicating an acceptable therapeutic index with respect to hERG. Similarly, across seven human CYP450 enzymes, weak inhibition

was observed with IC_{50} for CYP3A4 estimated as ≥ 10 μM, and CYP2C8 showing 84% inhibition at 10 μM. In the *in vivo* mouse PK studies (Table 5), compound **27** had moderate clearance and acceptable volume of distribution (1.49 L/kg). The elimination half-life was 2.9 h and the oral bioavailability was 49%.

Table 4. Properties of **27** (HCl salt) *in vitro*

Solubility (PBS, pH 7.4, μM)[a]	hERG inhibition IC_{50} (μM)[b]	CYP inhibition (% @ 10μM)[c]	Metabolic Stability in human hepatocytes[d]	Protein binding (%)[e]
73.43	29.258	1A2 (2.29) 2A6 (7.47) 2C8 (84.23) 2C9 (30.74) 2C19 (31.79) 2D6 (14.50) 3A4 (50.61)	*In vitro* $T_{1/2}$ (min): 202.00 *In vitro* Cl_{int} (μL/min/10^6 cells): 6.86	97.98

[a]Kinetic solubility. For reference, the solubility of progesterone is 22.37 μM. [b]HEK293 cell line stably transfected with hERG gene is employed and dofetilide is used as a positive control (IC_{50} = 0.015 μM). [c]For reference, the inhibitory percentage of subtypes of CYP by known inhibitors at 10 μM are CYP1A2 (phenacetin, 83.68), CYP2A6 (coumarin, 98.19), CYP2C8 (paclitaxel, 80.89), CYP2C9 (tolbutamide, 87.68), CYP2C19 (mephenytoin, 96.83), CYP2D6 (dextromethorphan, 94.37) and CYP3A4 (midazolam, 99.46). [d]For reference, *In vitro* $T_{1/2}$ and *In vitro* Cl_{int} of verapamil is 26.19 min and 52.91 μL/min/10^6 cells respectively. [e]For reference, protein binding of ketoconazole is 98.35%.

Early readouts from clinical trials showed that oncogene downregulation by BET inhibitors occurs within three hours of drug administration and suggests that a continuous exposure to BET inhibitor is not required to maintain oncogene suppression and antiproliferative activity.[60-62] Furthermore, it is widely believed that toxicity associated with BET inhibitors is on-target and therefore continuous exposure may be undesirable.[63, 64] To test if short-term exposure to **27** on would maintain efficacy, we conducted a cell culture experiment with drug washout , showing that the efficacy of compound **27** was only slightly diminished (right-shfted; Fig. 5G). Comparison was

made in washout experiments with BET inhibitors **1** and **5**, showing similar reduced efficacy after washout, with **1** showing no significant efficacy at < 300 nM. Although this cell culture model has limitations, the results suggest short-term exposure to **27** preserves efficacy.

Thrombocytopenia is a potential dose-limiting toxicity associated with anti-cancer agents including BET inhibitors. Several publications of BET inhibitor clinical trials have reported human thrombocytopenia readouts: for example, CPI-0610 was well tolerated, with the principal toxicity being reversible, non-cumulative, dose-dependent thrombocytopenia.[65] In contrast, there are few reports of *preclinical* analysis of thromombocytopenia induced by BET inhibitors. However, platelet data on BET inhibitor, **3** (3 mg/kg/day) has very recently been published in rats, which provides a validated model of clinical, dose-limiting thrombocytopenia.[64] Therefore, we measured platelet count in rats administered **27** p.o. at a dose (25 mg/kg/day) in rats equivalent to the dose in mice used in the xenograft study, compared rats administered compound **3**. The dose and response to **3** was comparable to the published report,[64, 66] and after 10 days of drug treatment, the platelet count was significantly reduced in the **3** treatment group compared to the **27** treatment group (Fig. 6A). The mean platelet count in rats before drug administration was 1088 ± 77 x 10^9 cells/mL compared to 173 ± 69 x 10^9 cells/mL and 1052 ± 85 x 10^9 cells/mL 10 days after treatment initiation for **3** and **27**, respectively.

The reported half-life for **3** varies across species (mice 2.7 h, rat 2.6 h, human 25.1 hours). Thus, although we observed a promising, neutral effect of **27** on platelet numbers in rats, it is impossible to attribute this to the somewhat short half-life of **27.** Although **27** was observed to have favorable characteristics in human liver hepatocytes and CYP450 inhibition, the extrapolation of ADME data in mice and platelet count in rats cannot be used to predict safety in humans without further ADMET studies.

Table 5. Pharmacokinetic Properties of **27** (HCl salt) in mice[a]

Route/dose	$T_{1/2}$ (h)	T_{max} (h)	C_{max} (ng/mL)	AUC (h*ng/mL)	Cl (mL/min/kg)	V_{ss} (L/kg)	*F* (%)
IV 5 mg/kg	1.95	-	-	3208	25.5	1.49	-
PO 30 mg/kg	2.90	0.25	3090	9598	-		49.0

[a]PK parameters from measurement of plasma concentration of **27** (p.o. 30 mg/kg) measured at 0.25, 0.5, 1, 2, 4, 8 and 24 h (N=3) in a formulation of 30% PEG400/70% HP-β-CD (10% in water) (v/v).

Compound **27** was studied in the tamoxifen-resistant MCF-7:TAM1 cell-derived xenograft model in nude mice. This tumorigenic cell line has been used to validate oral SERDs that have progressed to clinical trials in women with ER+ advanced breast cancer (ABC) who have failed on endocrine therapy.[67] Compound **27** (50 mg/kg/day p.o.) was compared with tamoxifen (1.5 mg/mouse/day p.o.), and vehicle control. The BET inhibitor significantly inhibited growth of tumors (Fig. 6B) without any effect on animal body weight (Fig. 6C). At day 28 of treatment, tamoxifen had no significant effect on tumor growth relative to the vehicle control group, whereas **27** significantly inhibited tumor growth relative to both the tamoxifen group and the vehicle control group (tumor growth inhibition: 106%, Fig. 6D). In addition to no loss of weight, no overt signs of toxicity were observed for once daily administration of **27** for 4 weeks. In a follow-up xenograft study, we compared **27** (30 mg/kg, p.o.) with **2** (50 mg/kg, p.o.) in combination with fulvestrant (s.c. 1 mg/mouse/week). Compound **2** was selected as an appropriate comparison, because it is the only BET inhibitor in clinical trials for advanced breast cancer in combination with fulvestrant (NCT02964507).[68] Both compounds showed comparable and significant tumor growth inhibition versus the vehicle control group (Fig. 6E).

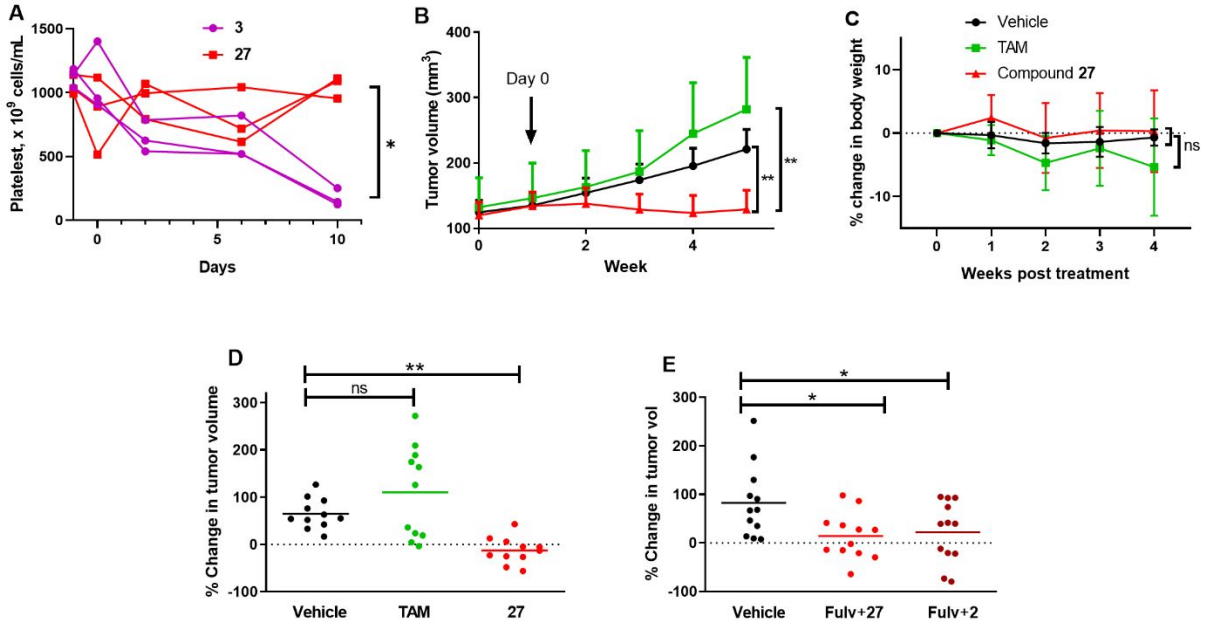


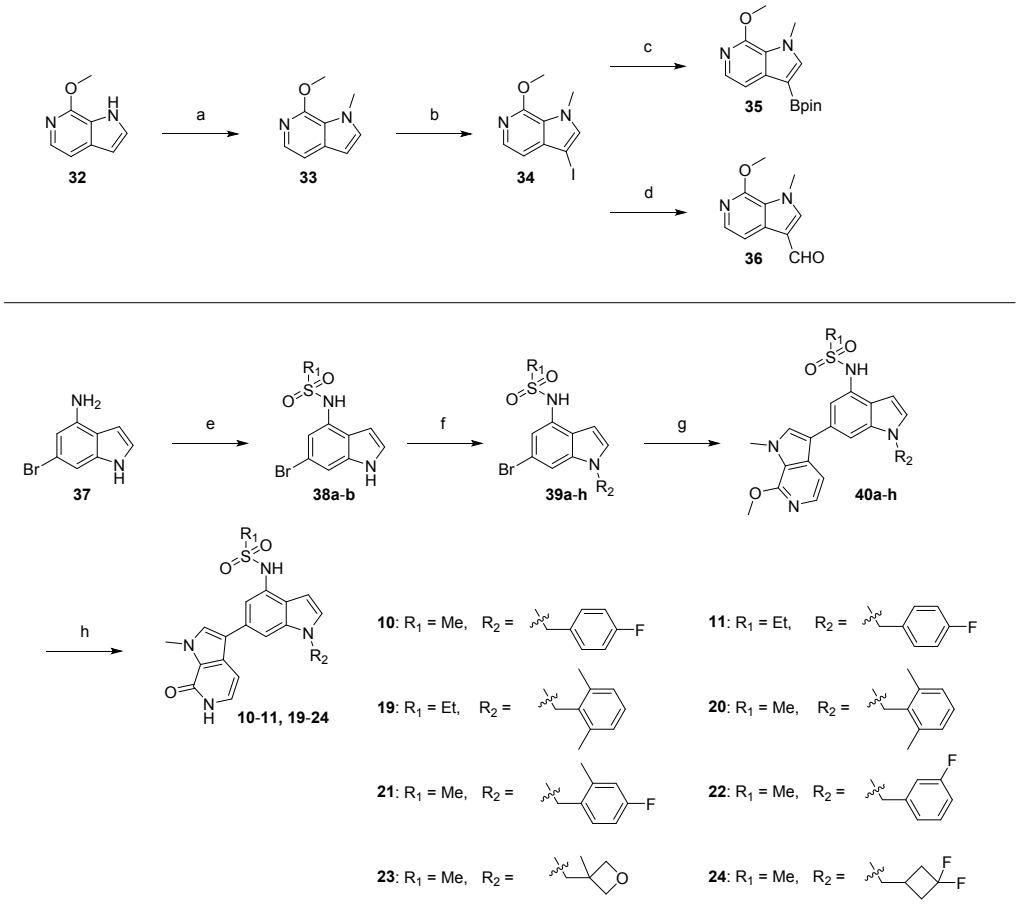
Figure 6. Platelet count in rats and inhibition of tumor growth in mice of tamoxifen-resistant MCF7:TAM1 cell-derived xenografts. A) Platelets counts from rats administered **27** (25 mg/kg) or **3** (3 mg/kg) starting at Day 0 showing data for individual rats (n=3): $p < 0.05$ at Day 10 by two-tailed paired t-test. B) Tumors grown to 0.32 cm^2 were treated with vehicle, TAM (1.5 mg/mouse/day), or compound **27** (HCl salt) (50 mg/kg/day) by daily oral gavage. Data show mean and S.D. C) Average body weight changes compared to pretreatment. D) Changes in individual tumor size at 4 weeks. E) Low dose fulvestrant (1 mg/mouse/week by s.c.) in combination with **2** 50 mg/kg/day (formulated in 0.5% (w/v) HPMC/TWEEN80 1% in water[69]) or **27** 30 mg/kg/day (formulated in 10% PEG400/90% HP-β-CD (10%)). Changes of individual tumor size at 4 weeks were plotted: data show mean and S.D. with one-way ANOVA analysis with Dunnett's multiple comparisons test.

Chemistry

The convergent synthesis of pyrrolopyridone BET inhibitors was mainly based on a Suzuki−Miyaura cross coupling reaction to construct a C-C bond between pyrrolopyridone moiety and the substitued indole moiety. The general synthesis of **10**-**11** and **19**-**24** is summarized in Scheme 1. The two components of the Suzuki−Miyaura coupling reaction were derived from commercially available compounds **32** and 6-bromo-1*H*-indol-4-amine (**37**). **32** was converted to **34** via a two-step sequence involving N-methylation followed by iodination of the 3-pyrrolo position. Iodo-substitued pyrrolopyridone, **34**, was then transformed to the organoboronic ester **35** in 65% yield by using one-pot lithiation–borylation reaction. Aryl bromide, the other component of the

Suzuki−Miyaura reaction, is prepared through a sequence of reactions starting from selective sulfonamidation of **37** to give **38a-b**, followed by S_N2 reaction with different benzyl or alkyl bromides to afford the corresponding **39a-h**. Suzuki−Miyaura coupling of precursor bromides with **35** using Pd(dppf)Cl_2-DCM as the catalyst gave **40a-h**. A mild condition for deprotection of the methyl ether using TMSCl and NaI generated the final products in near-quantitative yield.

Scheme 1. Synthetic route of Compound **10**-**11**, **19**-**24**

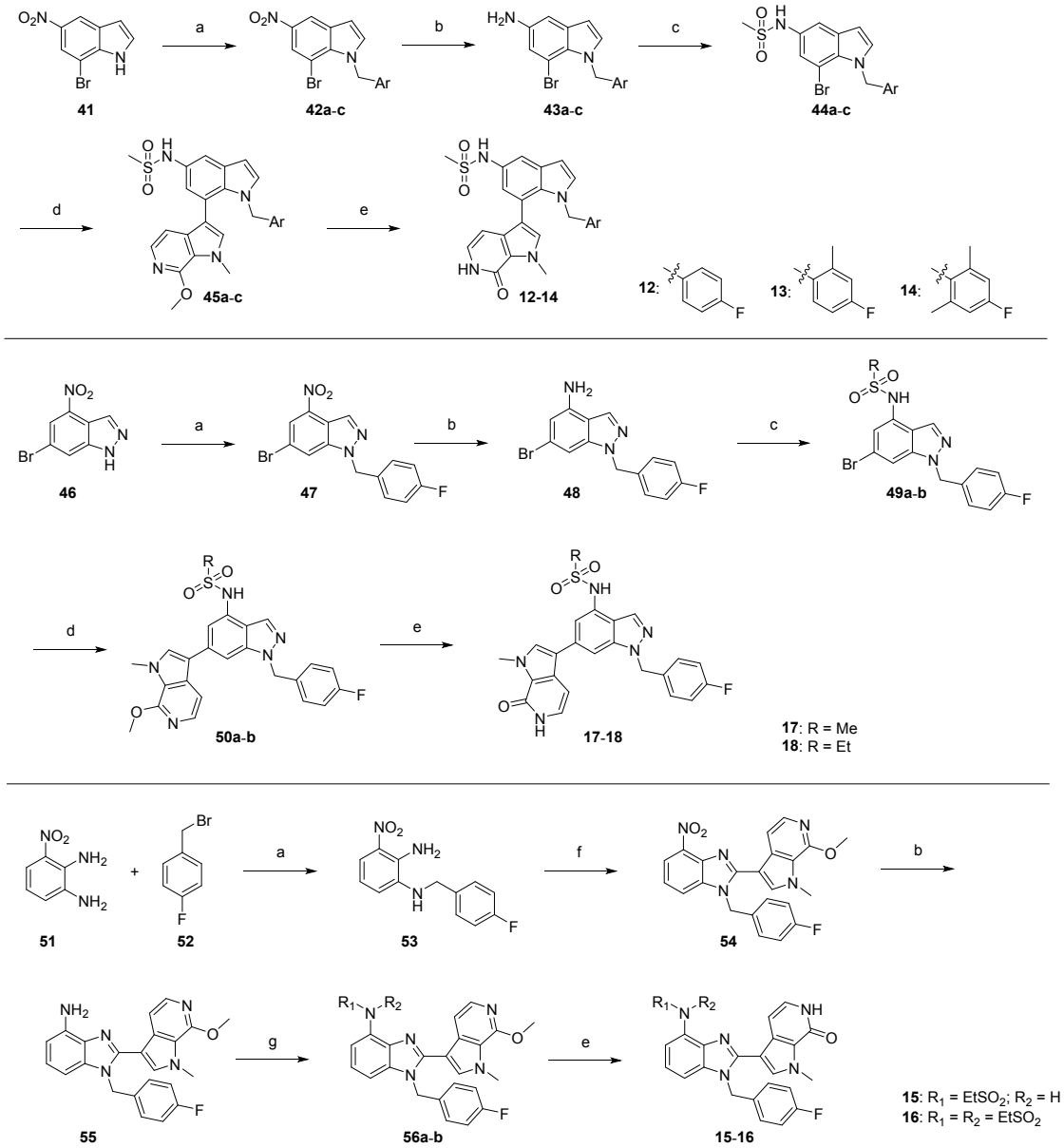


Reagents and conditions: (a) MeI, 60% NaH, DMF, RT, 1 h; (b) NIS, DMF, RT, 12 h, 92% over two steps; (c) *n*-BuLi, 2-isopropoxy-4,4,5,5-tetramethyl-1,3,2-dioxaborolane, THF, -78 °C to RT, overnight, 65%; (d) *n*-BuLi, DMF, THF, -78 °C, 2 h, 73%; (e) R_1SO_2Cl, pyridine, RT, 2 h, 75 - 96%; (f) R_2Br, K_2CO_3, DMF, 50 °C, 4 - 16 h, 66 - 92%; (g) **35**, Pd(dppf)Cl_2-DCM, K_3PO_4, dioxane/H_2O, 6 - 24 h, 90 °C, 77 - 94%; (h) TMSCl, NaI, MeCN/H_2O, 3 h, 50 °C, 82 - 96%.

The 7-indole pyrrolopyridone (**12**-**14**) and indazole derivatives (**17**-**18**) were prepared by using an analogous approach outlined in Scheme 2. Alkylation with the appropriate aryl bromide generated **42a**-**c** and **47**, followed by reduction of nitro group to provide anilines. Subsequent sulfonamidation of **43a**-**c** and **48** afforded corresponding Suzuki-Miyaura precusor bromides, which were then coupled with **35**. Compounds **12**-**14** and **17**-**18** were produced by deprotection of the corresponding methyl ethers.

Preparation of benzimidazole analogs, **15**-**16**, took a more linear synthesis because direct C-C bond construction between benzimidazole and pyrrolopyridone was not achieved with standard conditions. The 3-nitro-benzene-1,2-diamine, **51**, was treated with 4-fluorobenzyl bromide **52** to give the mono N-alkylation product **53**. The key oxidative cyclization of **53** with aldehyde **36** (prepared from **35** through formylation shown in Scheme 1) was achieved by heating with $NaBO_3$ in AcOH to yield **54**. Reduction of nitro group afforded the corresponding aniline **55**, which was further derivatized with ethanesulfonyl chloride to provide **56a**-**b**. Demethylation led to the final products **15**-**16**.

Scheme 2. Synthetic route of Compound **12**-**18**



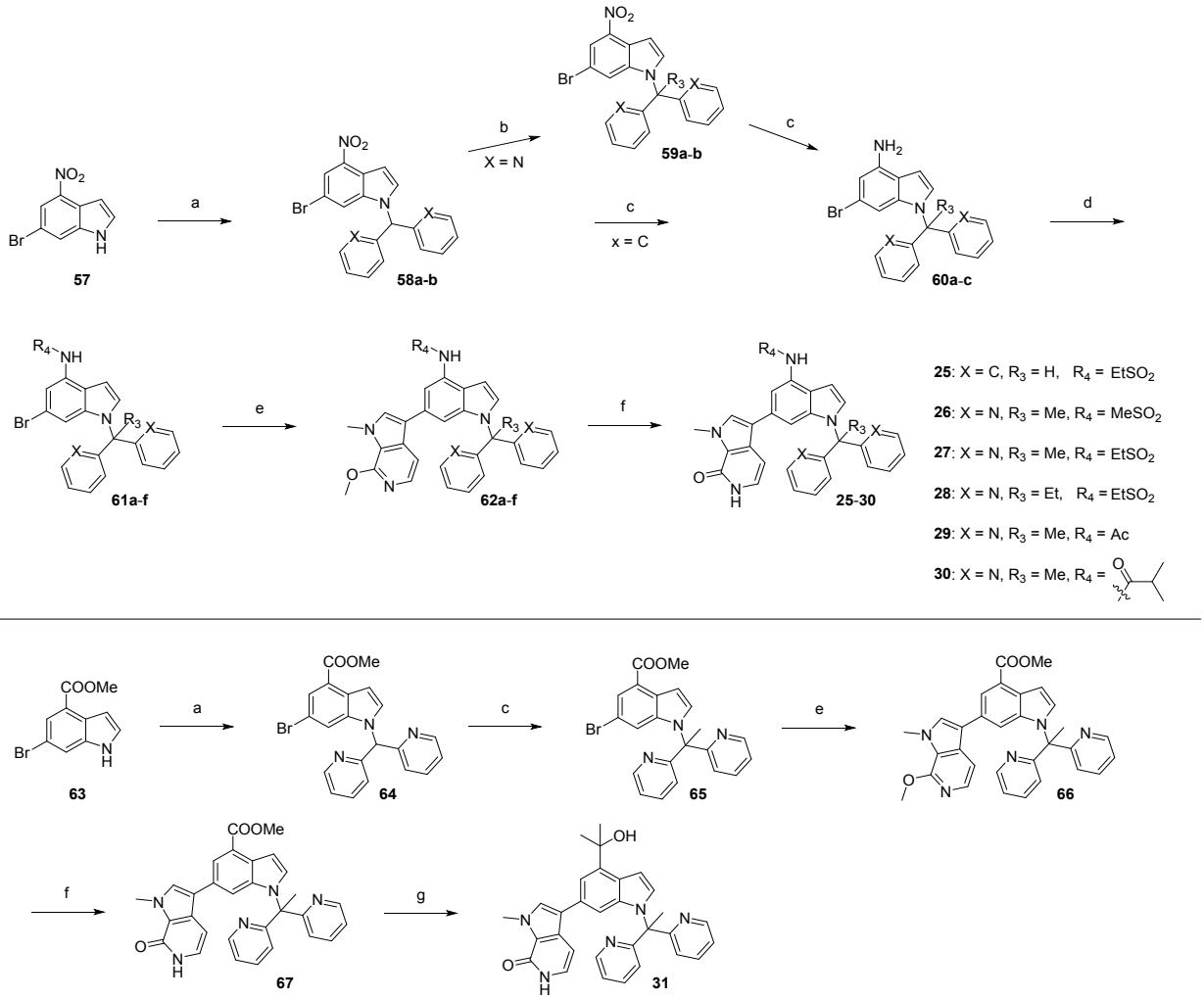
Reagents and conditions: (a) aryl bromide, K_2CO_3, DMF, RT, 2 h, 75 - 96%; (b) Fe, saturated NH_4Cl (aq.), EtOH, reflux, 2 h, 66 - 92%; (c) sulfonyl chloride, pyridine, RT, 1 - 5 h, 83 - 90%; (d) **35**, Pd(dppf)Cl_2-DCM, K_3PO_4, dioxane/H_2O, 6 - 24 h, 90 °C, 75 - 95%; (e) TMSCl, NaI, MeCN/H_2O, 2 h, 50 °C, 87 - 95%; (f) **36**, AcOH, $NaBO_3$, 50 °C, 1 h, 67%; (g) ethanesulfonyl chloride, triethylamine, DCM, RT, 3 h, 83 - 90%.

Bipyridine analogs, **25**-**30**, were prepared by using a convergent strategy similar to that depicted in Scheme 3. Treatment of **57** with 2,2'-(bromomethylene)dipyridine afforded **58a-b**, which was converted to **59a**-**b** in the presence of MeI and *t*-BuOK. It is noted that *t*-BuOK should

be added to the THF solution of MeI and substrate **58** to prevent unwanted initial formation of the debenzylation product. Reduction of nitro group using Fe/NH_4Cl system gave anilines **60a**-**c**, which then underwent acylation or sulfonylation to afford **61a**-**f**. With both boronic ester **35** and bromide **61c**, we directed our attention to optimize the Suzuki−Miyaura coupling reaction. A variety of conditions were explored, and we identified Pd(dppf)Cl_2-DCM as the optimal palladium catalyst with K_3PO_4 as the base in 1,4-dioxane/water (4:1, V/V) as solvent, at 90 °C (table S2). Demethylation gave the final products **25**-**30** in near-quantitative yield. Gram-scale synthesis of **27** was achieved through this same route.

Synthesis of compound **31** also used the Suzuki−Miyaura coupling reaction by starting from the commercially available compound **63**. Substitution followed by methylation of **64** furnished **65**, which was then coupled with **35** to yield **66**. Acid sensitive tertiary alcohol group forced us to carry out TMSCl-mediated deprotection first before it was introduced into the scaffold. Demethylation provided facile access to compound **67** without further purification. The final tertiary alcohol derivative, **31**, was obtained by treating **67** with methyllithium in THF at -42 °C.

Scheme 3. Synthetic route of Compound **25**-**30**



Reagents and conditions: (a) 2,2'-(bromomethylene)dipyridine or bromodiphenylmethane, K_2CO_3, DMF, 50 °C, 5 - 16 h, 72 - 91%; (b) R_3I, *t*-BuOK, THF, RT, 2 h, 73 - 93%; (c) Fe, saturated NH_4Cl (aq.), EtOH, reflux, 2 h; (d) sulfonyl chloride or acyl chloride, pyridine, RT, 1 - 5 h, 83 - 90%; (e) **35**, Pd(dppf)Cl_2-DCM, K_3PO_4, dioxane/H_2O, 90 °C, 6 - 24 h, 66 - 93%; (f) TMSCl, NaI, MeCN/H_2O, 50 °C, 2 h, 85 - 94%; (g) MeLi, THF, -42 °C, 2 h, 55%.

Discussion

ER+ breast cancer is a transcriptionally-driven cancer dependent on ER signaling.[22] Endocrine therapy using tamoxifen and AIs attempts to block ER signaling, either by selective modulation/antagonism of ER in breast tissues, or by inhibiting production of circulating estrogens. However, endocrine resistance occurs in 50% of breast cancer patients. The SERD, fulvestrant,

is currently the most efficacious ER-targeted therapy in metastatic disease, causing antagonism and loss of ER through proteasomal degradation. Together with the emergence of CDK4/6 inhibitors, the increasing use of fulvestrant has rapidly changed the standard of care for advanced and metastatic ER+ breast cancer. Resistance to fulvestrant and CDK4/6 inhibitors is an emerging clinical phenomenon. We envisioned that targeting of ER-driven transcription, via inhibition of bromodomain proteins, essential activating components of the ER transcriptional complex, would provide an alternative therapeutic approach to ER+ breast cancer.

Epigenetic dysregulation is a hallmark of cancer.[70] The discovery of the benchmark BET inhibitor, JQ1, provides the foundation for regulation of oncogene expression through "readers" of histone marks. The combination of a BRD-targeting BET inhibitor with a SERD would afford a more robust ablation of ER-driven transcription and circumvent resistance by attacking two different components of the ER transcriptional complex. The first generation BET inhibitor, **2**, has entered breast cancer clinical trials in combination with fulvestrant (NCT02964507). Herein, we report a novel BET inhibitor, **27**, with high binding affinity (K_d = 0.29 nM) and cellular potency (IC_{50} = 0.9 nM and >100 fold over JQ1). In two orthotopic xenograft models of endocrine-resistant ER+ breast cancer, oral administration of **27** significantly inhibited tumor growth, either as monotherapy or compared with **2** in combination with fulvestrant.

Most importantly, **27** is the first BET inhibitor to undergo iterative optimization using an ER+ endocrine-resistant breast cancer cell line, MCF-7:CFR that is also resistant to fulvestrant. This approach strongly supports the ability of **27** in combination with fulvestrant to block cancer cell escape pathways normally associated with fulvestrant resistance. Moreover, **27** maintained high antiproliferative potency in a range of cell lines modeling resistance to current therapies, including *ESR1* mutations and resistance to the leading CDK4/6 inhibitor, palbociclib. Iterative optimization was also guided by measuring BRD4-BD1 binding affinity by TR-FRET. The excellent

correlation between binding affinity and cellular potency across the 23 BET inhibitors discovered herein, support BRD4 as a primary target eliciting the observed antiproliferative phenotype.

The mechanisms by which **27** exerted antiproliferative activity in fulvestrant-resistant MCF-7:CFR cells was analyzed by RNA seq. These mechanisms include downregulation of oncogenes and up-regulation of tumor suppressor genes. Consistent with an early report on BET inhibitors, **27** downregulated components of both oncogenic MYC signaling, and cell cycle signaling.[26] The transcriptomic response to **27** was enhanced and differentiated relative to **1**, including a dramatic downregulation of anti-apoptotic and upregulation of apoptotic genes. These observations in a cell line resistant to endocrine therapy and fulvestrant provide a solid mechanistic basis for the further development of **27,** and BET inhibitors in general, for metastatic ER+ breast cancer.

Optimization was structure-based, driven by pyrrolopyridone and sulfonamide groups forming multiple hydrogen bonds with BRD4. The high potency of **27** was the result of adding a second pyridine group to the benzylic position leading to a novel binding interaction of this ring with the “leucine clamp” (L92 and L94), which greatly enhanced the binding affinity to BRD4. The unique approach and the rationale for high potency were supported by obtaining a crystal structure of **27** bound to BRD4. The increased affinity gained from the “leucine clamp” interaction provides a platform for further elaboration of the core of **27** to obtain domain-selective BET inhibitors and PROTACs, whilst maintain good affinity and potency (these will be the subject of future reports).

Although early clinical trials have revealed the activity of BET inhibitors in both hematological malignancies and solid tumors,[45] the clinical development of BET inhibitors is complicated by the roles of various BET proteins in multiple gene regulatory networks and signaling pathways. Many BET inhibitors in clinical development have suffered from reversible, dose-limiting thrombocytopenia in clinical trials.[6364] Evidence suggests that oncogene

downregulation is induced by BET inhibitors within 3 h of administration.[60-62] Thereofre, thrombocytopenia may be resolved by a 'dosing holiday' to allow systemic recovery,[65] or may be mitigated with supportive care agents such as romiplostim.[71] The PK characteristics of each BET inhibitor, in addition to selectivity and potency, therefore, become determinants of both efficacy and tolerability. Although further studies are needed, *in vitro* human and *in vivo* mouse ADME properties, combined with a effect on platelet count in rats, support the further development of **27** as a new targeted therapeutic in ER+ breast cancer; in particular, given efficacy in cancer cells resistant to fulvestrant and palbociclib.

Conclusion

In summary, a family of novel, orally bioavailable pyrrolopyridone BET inhibitors was discovered, culminating in **27** that displayed high selectivity and affinity for BRD proteins and high antiproliferative potency in tamoxifen, fulvestrant, and palbociclib-resistant and *ESR1* mutant breast cancer cell cultures. Data support a mechanism of action in fulvestrant-resistant cells mediated by binding to BRD4 with antiproliferative actions driven by downregulation of MYC and cylinD-cdk4 oncogenes and a significant shift from anti- to pro-apoptotic gene expression. The high potency and affinity of **27** for BRD4 was amply explained by novel interactions within a new co-crystal structure of **27.** ADMET and PK parameters showing oral bioavailability supported tumor xenograft studies that demonstrated growth inhibition of tamoxifen-resistant breast tumors in mice, with no body weight loss or overt toxicity. Similarly, studies in rats showed attenuated thrombocytopenia risk. Collectively, our data suggest that **27** is representative of a novel class of BET inhibitors with potential in cancer therapy.

EXPERIMENTAL SECTION

Cell Lines and Culture Conditions. A detailed derivatization of MCF-7 cell lines is summarized in Supporting Fig. S1. The parental, tamoxifen-resistant cell line, MCF7:5C, was derived from a single cell clone by the Jordan group at Northwestern University as previously described.[72, 73] These cells were maintained in phenol red-free RPMI 1640 medium supplemented with 10% charcoal-dextran treated FBS, 1% antibiotic-antimycotic, and 1% glutamax at 37 °C and 5% CO_2. The MCF7:CFR cell line was generated through long-term exposure to increasing concentrations of fulvestrant (200 nM - 1 μM) until resistance was established (>25 passages). MCF7:CFR cells are maintained in fulvestrant (200 nM) and phenol red-free, stripped RPMI-1640. MCF7:ws8 cells were cultured in RPMI-1640 media with 10% FBS, 1% non-essential amino acids (NEAA), antibiotic-antimitotic (AbAm), and Glutamax, and 10 ng/ml insulin. MCF7:TAM1 cells were maintained in the same media as MCF7:5C with the addition of 100 nM 4OHT. MCF7:CPR cells were generated with increasing concentrations of CDK4/6 inhibitor (100 nM to 1000 nM), palbociclib, and maintained in the same media as MCF7:5C with the addition of 200 nM palbociclib. T47D parental and T47D/TDG are gifts from David Shapiro's lab at University of Illinois at Urbana-Champaign.[74]

Cell Growth Assay. Compounds **1**-**6** were purchased from MedChemExpress. MCF-7:CFR cells were grown in phenol red-free RPMI 1640 medium and 10% charcoal-dextran treated FBS for 2 days prior to each experiment. One day before the treatment, cells were seeded in a 96-well clear, flat bottom plate at a density of 2000 cells/well and treated with either 0.1% (v/v) DMSO or compounds of interest (0.1 nM - 10 μM). All compounds were dissolved in DMSO and added to the medium at a final 1:1000 dilution. Cell number was monitored daily via Celigo imaging system (Nexcelom Bioscience). On day 5, media was removed, and cells were quantified by using Hoechst 33342 dye staining and quantified by Celigo. Cell growth assay in other cells were performed in a similar manner with corresponding medium. The 3-hour treatment and

washout study in MCF-7:CFR cell cultures was carried out in similar fashion except switching treatment medium after 3 hours to normal growth medium.

3D-spheroid growth assay. MCF7:CFR cells were plated at a density of 1000 cells/well in Corning® 96-well black, clear round bottom, ultra-low attachment spheroid microplates and grown in the absence of treatment for 24 hours. Spheroids were then treated with 2X treatment media following the removal of half of the media (100 µL) from each well. Treatment was repeated every 2-3 days for 14 days. CellTiter-Glo® 3D Cell Viability Assay protocol was used to determine growth inhibition of the spheroids, as per manufacturer's instructions. Luminescence signal was read using a Synergy Neo (Biotek). Data was normalized to blank (media with CellTiter Glo 3D reagent).

Time-resolved fluorescence energy transfer (TR-FRET) bromodomain binding assay. The binding affinity of our chemical library was assayed measured using Cayman BRD4 bromodomain 1 TR-FRET assay kit (cat #: 600520) as per manufacturer's instructions. Briefly, 5 µL of tested compound and 10 µL of BRD4 BD1 europium chelate were added to Corning® 384-well flat, clear plate and pre-incubated for 15 min. Subsequently, 5 µL of the BRD4 bromodomain 1 ligand/APC acceptor mixture was added to each tested well. The plate was then sealed and incubated at room temperature in the dark for 1 hour. The TR-FRET signal was read using a Synergy Neo (Biotek) by measuring the dual emissions at 620 nm and 670 nm using an excitation at 340 nm. The data is analyzed using TR-FRET ratio (670 nm/620 nm emission) and normalized to DMSO as 100%.

Protein purification. The gene of human BRD4-BD1 (residues 44-168) with a His_6-SUMO-tag at the N-terminus were codon-optimized, synthesized (Genscript), and cloned into a pET15b vector between XhoI and BamHI. The plasmid was transformed into *E. coli* BL21(DE3) cell for protein expression. One liter of BL21(DE3) cells containing the recombinant BRD4-BD1 gene were grown to an OD_{600} of 0.65-0.7 at 37 °C in LB medium containing 100 µg/mL ampicillin.

The cells were then induced with 0.5 mM IPTG for 14 hours at 25 °C. Harvested cells were lysed by sonication in lysis buffer (1 mg/mL lysozyme, 0.01% TritonX-100, 0.025 mg/mL DNase I, and one tablet of EDTA-free protease inhibitor cocktail in buffer A: 50 mM HEPES, pH 8.0, 500 mM NaCl, 5 mM Imidazole, 5% glycerol, and 0.5 mM TCEP). The His_6-SUMO fused BRD4-BD1 was purified by a two-step protocol, a HisTrap HP column (GE Healthcare) with a stepwise gradient of buffer B (50 mM HEPES, pH 8.0, 500 mM NaCl, 500 mM imidazole, 5% glycerol, and 0.5 mM TCEP) and a 16/60 Superdex 75 size exclusion column with buffer C (50 mM HEPES, pH 8.0 , 150 mM NaCl, 5% glycerol, and 0.5 mM TCEP). The His_6-SUMO tag was cleaved by 2 units/mL of SUMO protease for 2 hours at room temperature, producing a BRD4-BD1 with only two extra residues (LE) at the N-terminus. Finally, digested sample was again loaded onto the HisTrap column to remove SUMO protease, cleaved His-SUMO tags, and uncleaved His_6-SUMO fused BRD4-BD1. Purified BRD4-BD1 protein with approximately 95% purity was either used to set up crystals or flash frozen in liquid nitrogen for storage at -80 °C.

Protein Crystallography. Crystals of BRD4(BD1) complexed with **27** were grown by hanging drop vapor diffusion at 4°C. Prior to crystallization, 10 mg/mL BRD4(BD1) was incubated with 2.5 mM **27** for 30 min at room temperature and then centrifuged to remove precipitate. Crystals of the complex were grown by mixing 2 μL of BRD4(BD1): **27** with 1-2 μL of reservoir solution containing 16-20% PEG 4000, 0.2 M Li_2SO_4, and 0.1 M Tris-HCl, pH 8.5. Crystals were cryo-preserved by soaking in mother liquor containing 10% glycerol and 200 μM **27** and frozen directly in the liquid nitrogen stream of the crystal mount prior to data collection. Data were collected on a MAR300 detector at 0.979 Å at the Life Sciences Collaborative Access Team beamline 21-ID-F at the Advanced Photon Source, Argonne National Laboratory. Data indexing, integration, and scaling were performed using XDS[75] (**1**), and phases were determined by molecular replacement using Molrep[76] (**2**) and a BRD4(BD1) structure (PDB entry: 3MXF) as search model. Rigid body refinement followed by iterative rounds of restrained refinement and

model building were performed with CCP4i[77] (**3**) modules Refmac5[78] (**4**) and Coot[79] (**5**). The coordinates and structure factors have been deposited with PDB accession code: 6P05.

BROMOscan Bromodomain Profiling. BROMOscan bromodomain profiling was provided by DiscoverX Corp. K_d of test compounds with DNA-tagged bromodomains was determined through binding against a proprietary reference immobilized ligand.

RNA seq analysis. MCF-7:CFR cells were treated with DMSO (0.01%), **1** (100 nM) and **27** (100 nM) for 24 hours. RNA was extracted using the RNeasy Plus Mini Kit (Qiagen) as per the manufacturer's directions. QC, sequence and data analysis were performed by Novogene.

Pharmacokinetics Studies. The Animal Care and Use Committee of the University of Illinois at Chicago approved all the procedures involving animals. Kinetic solubility, hERG inhibition, hepatocyte stability, and human human CYP450 enzymes inhibition were performed by Pharmaron Inc. Rigorous mouse PK profiling was also conducted by Pharmaron Inc. Briefly, HCl salt of **27** was formulated in 30% PEG400 and 70% "10% HP-β-CD in water" (v/v) in a concentration of 1 mg/mL for IV injection and 3 mg/mL for oral gavage. Blood samples (100 μL) were collected from CD1 mice at 0, 0.0833, 0.25, 0.5, 1, 2, 4, 8, and 24 h for IV, and 0, 0.25, 0.5, 1, 2, 4, 8, and 24 h for p.o. (n = 3 for each time point). Plasma concentrations of the compounds were determined by AB API 6500 QTRAP LC/MS/MS instrument developed and validated for this study.

Animal Experiments. The Animal Care and Use Committee of the University of Illinois at Chicago approved all the procedures involving animals. For the mono therapy animal study, MCF-7:Tam1 tumors were established in 4-6 week old female ovariectomized athymic nude mice (Harlan Laboratories; Strains: nude/Foxn1) with a daily tamoxifen treatment of 1.5 mg/mouse as previously described.[80-82] **27** was administered by gavage at a dose of 50 mg/kg/day for 4 weeks in a formulation of 10% PEG400 and 90% HP-β-CD (10% in water) (v/v). For the combinational

therapy animal study, MCF-7:Tam1 tumors were established in 4-6 week old female ovariectomized athymic nude mice (Harlan Laboratories; Strains: nude/Foxn1) with a daily tamoxifen treatment of 1.5 mg/mouse as previously described.[80-82] **27** was administered by gavage at a dose of 30 mg/kg/day for 4 weeks in a formulation of 10% PEG400 and 90% HP-β-CD (10% in water) (v/v). **2** was administered by gavage at a dose of 50 mg/kg/day for 4 weeks in a formulation of 0.5% (w/v) HPMC/TWEEN80 1% in water [69]. Tumor volume (TV) was determined weekly using Vernier calipers and calculated using the formula (length) × (width)^2 × π/6. Mean tumor area was plotted against time (in weeks) to monitor tumor growth. Tumor growth inhibition was calculated by formula, (1– ($TV_{tx\ Day\ 28}$ - $TV_{tx\ Initial}$)/ ($TV_{veh\ Day\ 28}$ - $TV_{veh\ Initial}$)) x 100%. For the thrombocytopenia study, 7-week-old male Sprague-Dawley rats were obtained from Taconic (Rensselaer, New York, USA) and housed two rats per cage. Water and food were provided *ad libitum*. One-week acclimation period was allowed before treatment started. During acclimation period blood was collected twice to determine the baseline platelet count. Drug treatment began at 8 weeks of age. All compounds were administrated by oral gavage (10 mL/kg) daily from 4:00 to 5:00 p.m. During treatment period, the blood was collected 72, 168 and 264 hrs after treatment initiation and platelets counted. Blood collection was performed from 11:00-12:00. For the blood collection, animals were anesthetized with 2.5% isoflurane inhalation and no more than 3% of estimated circulating blood volume has been withdrawn from tail vein using standard IACUC approved procedure for tail vein blood collection. Animal weight, body, and behavior were monitored daily.

General Chemical Experimental Information Unless otherwise specified, reactions were performed under an inert atmosphere of argon and monitored by thin-layer chromatography (TLC) and/or LCMS. All reagents were purchased from commercial suppliers and used as provided. Synthetic intermediates were purified using CombiFlash chromatography system on 230−400 mesh silica gel. ^{1}H and ^{13}C NMR spectra were obtained using Bruker DPX-400 or

AVANCE-400 spectrometer at 400 and 100 MHz, respectively. NMR chemical shifts were described in δ (ppm) using residual solvent peaks as standard (Chloroform-*d*, 7.26 ppm (^{1}H), 77.16 ppm (^{13}C); Methanol-d_4, 3.31 ppm (^{1}H), 49.00 ppm (^{13}C); DMSO-d_6, 2.50 ppm (^{1}H), 39.52 ppm (^{13}C); Acetone-d_6, 2.05 ppm (^{1}H), 29.84 ppm (^{13}C)). Data were reported in a format as follows: chemical shift, multiplicity (s = singlet, d = doublet, dd = doublet of doublet, t = triplet, q = quartet, br = broad, m = multiplet, abq = ab quartet), number of protons, and coupling constants. High resolution mass spectral data were measured in-house using a Shimadzu IT-TOF LC/MS for all final compounds. All compounds submitted for biological testing were confirmed to be ≥ 95% pure by analytical HPLC. Synthetic methods, spectral data, and HRMS for novel compounds are described in detail below.

7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridine (**33**). To a solution of **32** (2.96 g, 20 mmol) in DMF (50 mL) at 0 °C was added 60% NaH (0.96 g, 24 mmol) and methyl iodide (1.37 mL, 22 mmol). The reaction mixture was stirred at room temperature (RT) for 1 h. Water was then added to quench the reaction. The solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave **33** (3.21 g) without further purification.

3-iodo-7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridine (**34**). To a solution of **33** (from previous step) in DMF (60 mL) at RT was added NIS (4.95 g, 22 mmol) and then the mixture was stirred at RT for 12 h. Water was added to quench the reaction. The solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 3:1) to obtain **34** as a white solid (5.3 g, yield 92% over the two steps). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.76 (d, *J* = 5.7 Hz, 1H), 7.11 (s, 1H), 6.92 (d, *J* = 5.6 Hz, 1H), 4.08 (s, 3H), 4.06 (s, 3H). LCMS (m/z) $[M + H]^+$ = 289.0.

7-methoxy-1-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1*H*-pyrrolo[2,3-*c*]pyridine (**35**). To a solution of **34** (2.88 g, 10 mmol) in THF (20 mL) was added *n*-BuLi (12 mmol) at -78 °C. The rection mixture was stirred for 1 h, followed by addition of 2-isopropoxy-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (2.24 mL, 11 mmol) at -78 °C. The reaction mixture was allowed to warm to room temperatue and stirred for overnight. Upon the completion of the reaction monitored by TLC, water was added to quench the reaction. The solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 10:1) to obtain 1.87 g of white solid **35** with a 65% yield. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.75 (d, *J* = 5.6 Hz, 1H), 7.48 (d, *J* = 5.6 Hz, 1H), 7.44 (s, 1H), 4.07 (s, 3H), 4.04 (s, 3H), 1.34 (s, 12H). LCMS (m/z) $[M + H]^+$ = 289.2.

N-(6-bromo-1*H*-indol-4-yl)methanesulfonamide (**38a**). To a solution of **37** (4.22 g, 20 mmol) in pyridine (60 mL) at 0 °C was added MsCl (3.0 mL, 40 mmol). The reaction mixture was allowed to slowly warm to room temperatue. The solution was further stirred at RT for 2 h and then quenched by adding water. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. And then filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 1:1) to obtain **38a** (5.26 g, yield 91%) as a brown solid. ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.55 (s, 1H), 8.55 (s, 1H), 7.49 (dd, *J* = 1.6, 1.0 Hz, 1H), 7.38 (dd, *J* = 3.3, 2.4 Hz, 1H), 7.34 (d, *J* = 1.6 Hz, 1H), 6.88 (ddt, *J* = 3.2, 2.1, 1.3 Hz, 1H), 3.01 (d, *J* = 1.1 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 289.1.

N-(6-bromo-1*H*-indol-4-yl)ethanesulfonamide (**38b**). This compound was obtained by a procedure similar to the preparation of **38a** (860 mg, yield 95%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.33 (s, 1H), 7.41 (t, *J* = 1.3 Hz, 1H), 7.39 (d, *J* = 1.4 Hz, 1H), 7.22 (dd, *J* = 3.4, 2.4 Hz, 1H),

6.58 (s, 1H), 6.54 (dt, *J* = 2.9, 1.5 Hz, 1H), 3.19 (q, *J* = 7.4 Hz, 2H), 1.38 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 303.0.

N-(6-bromo-1-(4-fluorobenzyl)-1*H*-indol-4-yl)methanesulfonamide (**39a**). To a solution of **38a** (4.34 g, 15 mmol) in DMF (50 mL) at RT was added K_2CO_3 (3.11 g, 22.5 mmol) and **52** (2.24 mL, 18 mmol). The solution was stirred at 50 °C for 6 h, followed by quenching in water. The mixture was then extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 2:1) to obtain **39a** as a yellow solid (5.48 g, yield 92%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.37 (d, *J* = 1.4 Hz, 1H), 7.29 (t, *J* = 1.1 Hz, 1H), 7.11 (d, *J* = 3.3 Hz, 1H), 7.08 (dd, *J* = 8.6, 5.4 Hz, 2H), 7.05 – 6.98 (m, 2H), 6.65 (s, 1H), 6.54 (dd, *J* = 3.3, 0.9 Hz, 1H), 5.24 (s, 2H), 3.07 (s, 3H). LCMS (m/z) $[M + H]^+$ = 397.2.

N-(6-bromo-1-(4-fluorobenzyl)-1*H*-indol-4-yl)ethanesulfonamide (**39b**). This compound was obtained by a procedure similar to the preparation of **39a** (221 mg, yield 90%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.40 (s, 1H), 7.42 (t, *J* = 1.2 Hz, 1H), 7.21 – 7.15 (m, 2H), 7.12 (dd, *J* = 3.3, 2.4 Hz, 1H), 7.09 (d, *J* = 1.5 Hz, 1H), 6.93 – 6.87 (m, 2H), 6.43 (ddd, *J* = 3.2, 2.1, 1.0 Hz, 1H), 4.89 (s, 2H), 3.13 (q, *J* = 7.4 Hz, 2H), 1.43 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 411.0.

N-(6-bromo-1-(2,6-dimethylbenzyl)-1*H*-indol-4-yl)ethanesulfonamide (**39c**). This compound was obtained by a procedure similar to the preparation of **39a** (178 mg, yield 76%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 7.55 (dd, *J* = 1.6, 0.9 Hz, 1H), 7.31 (dd, *J* = 3.2, 2.5 Hz, 1H), 7.21 (d, *J* = 1.6 Hz, 1H), 6.89 (dd, *J* = 8.3, 6.6 Hz, 1H), 6.81 (d, *J* = 7.5 Hz, 2H), 6.55 (ddd, *J* = 3.1, 2.1, 0.9 Hz, 1H), 5.14 (s, 2H), 3.26 (q, *J* = 7.4 Hz, 2H), 2.27 (s, 6H), 1.37 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 421.1.

N-(6-bromo-1-(2,6-dimethylbenzyl)-1*H*-indol-4-yl)methanesulfonamide (**39d**). This compound was obtained by a procedure similar to the preparation of **39a** (95 mg, yield 75%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.28 – 7.23 (m, 1H), 7.20 – 7.14 (m, 3H), 6.60 (d, *J* = 1.5 Hz, 1H), 6.49 (d, *J* = 3.3 Hz, 1H), 6.30 (dd, *J* = 3.3, 0.9 Hz, 1H), 5.16 (s, 2H), 3.99 (s, 3H), 2.29 (s, 6H). LCMS (m/z) $[M + H]^+$ = 407.0.

N-(6-bromo-1-(4-fluoro-2-methylbenzyl)-1*H*-indol-4-yl)methanesulfonamide (**39e**). This compound was obtained by a procedure similar to the preparation of **39a** (113 mg, yield 84%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.38 (dd, *J* = 3.4, 1.5 Hz, 1H), 7.30 (t, *J* = 1.2 Hz, 1H), 6.95 (m, 2H), 6.83 (td, *J* = 8.4, 2.7 Hz, 1H), 6.78 – 6.71 (m, 2H), 6.53 (dd, *J* = 3.4, 0.9 Hz, 1H), 5.18 (s, 2H), 3.08 (s, 3H), 2.27 (s, 3H). LCMS (m/z) $[M + H]^+$ = 411.0.

N-(6-bromo-1-(3-fluorobenzyl)-1*H*-indol-4-yl)methanesulfonamide (**39f**). This compound was obtained by a procedure similar to the preparation of **39a** (89 mg, yield 92%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.53 (s, 1H), 7.34 (t, *J* = 1.2 Hz, 1H), 7.18 (td, *J* = 7.9, 5.8 Hz, 1H), 7.11 (d, *J* = 1.5 Hz, 1H), 7.06 (dd, *J* = 3.3, 2.4 Hz, 1H), 7.02 (dt, *J* = 7.7, 1.2 Hz, 1H), 6.98 (dt, *J* = 9.5, 2.2 Hz, 1H), 6.89 (tdd, *J* = 8.5, 2.7, 1.0 Hz, 1H), 6.43 (ddd, *J* = 3.2, 2.0, 1.0 Hz, 1H), 4.90 (s, 2H), 3.01 (s, 3H). LCMS (m/z) $[M + H]^+$ = 397.1.

N-(6-bromo-1-((3-methyloxetan-3-yl)methyl)-1*H*-indol-4-yl)methanesulfonamide (**39g**). This compound was obtained by a procedure similar to the preparation of **39a** (86 mg, yield 93%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.37 (s, 2H), 7.25 (s, 1H), 7.06 (d, *J* = 3.3 Hz, 1H), 6.62 (d, *J* = 3.2 Hz, 1H), 4.63 (d, *J* = 6.1 Hz, 2H), 4.41 (d, *J* = 6.1 Hz, 2H), 4.31 (s, 2H), 3.06 (s, 3H), 1.30 (s, 3H). LCMS (m/z) $[M + H]^+$ = 373.1.

N-(6-bromo-1-((3,3-difluorocyclobutyl)methyl)-1*H*-indol-4-yl)methanesulfonamide (**39h**). This compound was obtained by a procedure similar to the preparation of **39a** (158 mg, yield 96%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.52 (s, 1H), 7.37 (d, *J* = 1.5 Hz, 1H), 7.34 (t, *J* =

1.1 Hz, 1H), 7.06 (d, *J* = 3.3 Hz, 1H), 6.67 (dd, *J* = 3.3, 0.9 Hz, 1H), 4.17 (d, *J* = 5.9 Hz, 2H), 3.07 (s, 3H), 2.74 – 2.60 (m, 3H), 2.42 – 2.26 (m, 2H). LCMS (m/z) $[M + H]^+$ = 392.9.

N-(1-(4-fluorobenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**40a**). A mixture of **39a** (4.77 g, 12 mmol), Pd(dppf)Cl_2-DCM (490 mg, 0.6 mmol), **35** (4.16 g, 14.4 mmol), and K_3PO_4 (6.62 g, 31.2 mmol) in dioxane-water (25 mL, V/V, 4:1) was heated at 90 °C for 24 h. The mixture was then diluted with ethyl acetate and filtered to obtain a clear solution, which was subsequently washed three times with brine. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by flash column chromatography on silica gel (Hexanes-Acetone, 1:1) to afford **40a** as a grey solid (5.16 g, yield 90%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.47 (s, 1H), 7.67 (d, *J* = 5.8 Hz, 1H), 7.54 – 7.48 (m, 3H), 7.42 (d, *J* = 3.3 Hz, 1H), 7.33 (dd, *J* = 8.7, 5.6 Hz, 3H), 7.11 (t, *J* = 8.8 Hz, 2H), 6.94 – 6.88 (m, 1H), 5.51 (s, 2H), 4.11 (s, 3H), 4.05 (s, 3H), 3.02 (s, 3H). LCMS (m/z) $[M + H]^+$ = 479.2.

N-(1-(4-fluorobenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**40b**). This compound was obtained by a procedure similar to the preparation of **40a** (102 mg, yield 94%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.20 (d, *J* = 0.9 Hz, 1H), 7.77 (d, *J* = 5.8 Hz, 1H), 7.57 (s, 1H), 7.46 (d, *J* = 1.0 Hz, 1H), 7.27 (d, *J* = 1.5 Hz, 2H), 7.25 – 7.20 (m, 3H), 7.04 – 6.98 (m, 2H), 5.57 (s, 2H), 4.11 (s, 3H), 4.10 (s, 3H), 3.25 (q, *J* = 7.4 Hz, 2H), 1.41 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 493.1.

N-(1-(2,6-dimethylbenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**40c**). This compound was obtained by a procedure similar to the preparation of **40a** (98 mg, yield 83%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 7.71 (d, *J* = 5.7 Hz, 1H), 7.59 (dd, *J* = 3.8, 1.1 Hz, 2H), 7.57 (s, 1H), 7.42 (d, *J* = 5.7 Hz, 1H), 7.23 (dd, *J* = 8.4, 6.5 Hz, 1H), 7.16 (d, *J* = 7.5 Hz, 2H), 6.83 (d, *J* = 3.3 Hz, 1H), 6.78 (d, *J* = 3.3 Hz, 1H), 5.45 (s, 2H),

4.15 (s, 3H), 4.06 (d, *J* = 1.8 Hz, 3H), 3.13 (q, *J* = 7.4 Hz, 2H), 2.30 (s, 6H), 1.30 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 503.2.

N-(1-(2,6-dimethylbenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**40d**). This compound was obtained by a procedure similar to the preparation of **40a** (91 mg, yield 90%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.44 (s, 1H), 7.71 (dd, *J* = 5.7, 1.3 Hz, 1H), 7.62 (d, *J* = 1.2 Hz, 1H), 7.58 (s, 1H), 7.55 (d, *J* = 1.3 Hz, 1H), 7.41 (d, *J* = 5.7 Hz, 1H), 7.25 – 7.21 (m, 1H), 7.17 (s, 1H), 7.15 (s, 1H), 6.79 (s, 2H), 5.46 (s, 2H), 4.15 (d, *J* = 1.5 Hz, 3H), 4.07 (s, 3H), 3.01 (s, 3H), 2.30 (d, *J* = 2.2 Hz, 6H). LCMS (m/z) $[M + H]^+$ = 489.1.

N-(1-(4-fluoro-2-methylbenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**40e**). This compound was obtained by a procedure similar to the preparation of **40a** (75 mg, yield 77%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.74 (d, *J* = 5.7 Hz, 1H), 7.49 (d, *J* = 1.2 Hz, 1H), 7.31 (d, *J* = 1.0 Hz, 1H), 7.24 (d, *J* = 5.8 Hz, 1H), 7.22 (s, 1H), 7.00 (d, *J* = 3.3 Hz, 1H), 6.94 (dd, *J* = 9.5, 2.4 Hz, 1H), 6.87 – 6.82 (m, 2H), 6.63 (t, *J* = 2.5 Hz, 1H), 5.25 (s, 2H), 4.09 (s, 3H), 4.07 (s, 3H), 3.08 (s, 3H), 2.28 (s, 3H). LCMS (m/z) $[M + H]^+$ = 493.1.

N-(1-(3-fluorobenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**40f**). This compound was obtained by a procedure similar to the preparation of **40a** (104 mg, yield 84%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.96 (t, *J* = 2.3 Hz, 1H), 7.71 (d, *J* = 5.7 Hz, 1H), 7.39 (t, *J* = 1.1 Hz, 1H), 7.21 (d, *J* = 1.3 Hz, 1H), 7.20 – 7.14 (m, 3H), 7.11 – 7.03 (m, 2H), 7.00 (s, 1H), 6.91 – 6.85 (m, 1H), 6.50 (ddd, *J* = 3.1, 2.0, 1.0 Hz, 1H), 4.99 (s, 2H), 4.08 (s, 3H), 3.96 (s, 3H), 3.03 (s, 3H). LCMS (m/z) $[M + H]^+$ = 479.1.

N-(6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1-((3-methyloxetan-3-yl)methyl)-1*H*-indol-4-yl)methanesulfonamide (40g). This compound was obtained by a procedure similar to the preparation of **40a** (129 mg, yield 90%). ^{1}H NMR (400 MHz, Chloroform-*d*)

δ 7.78 (d, *J* = 5.7 Hz, 1H), 7.44 (d, *J* = 22.0 Hz, 2H), 7.39 (d, *J* = 5.7 Hz, 1H), 7.09 (d, *J* = 3.3 Hz, 1H), 6.72 (s, 1H), 6.56 (d, *J* = 3.3 Hz, 1H), 4.70 (d, *J* = 6.1 Hz, 2H), 4.44 (d, *J* = 6.1 Hz, 2H), 4.40 (s, 2H), 4.11 (s, 3H), 4.10 (s, 3H), 3.06 (s, 3H), 1.35 (s, 3H). LCMS (m/z) $[M + H]^+$ = 455.2.

N-(1-((3,3-difluorocyclobutyl)methyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (40h). This compound was obtained by a procedure similar to the preparation of **40a** (95 mg, yield 86%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.78 (d, *J* = 5.8 Hz, 1H), 7.59 (s, 1H), 7.53 – 7.49 (m, 1H), 7.43 (d, *J* = 5.8 Hz, 1H), 7.37 (s, 1H), 7.06 (d, *J* = 3.2 Hz, 1H), 6.68 (d, *J* = 3.2 Hz, 1H), 4.22 (d, *J* = 6.5 Hz, 2H), 4.10 (s, 3H), 4.06 (s, 3H), 3.06 (s, 3H), 2.75 – 2.63 (m, 3H), 2.41 – 2.29 (m, 2H). LCMS (m/z) $[M + H]^+$ = 475.2.

N-(1-(4-fluorobenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**10**). A mixture of **40a** (478 mg, 1 mmol), TMSCl (203 μL, 1.6 mmol), NaI (240 mg, 1.6 mmol), and H_2O (9 μL, 0.5 mmol) in acetonitrile (6 mL) was heated at 50 °C for 3 h. The reaction mixture was then cooled to RT, diluted with ethyl acetate, and washed three times with aqueous $NaHCO_3$. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by preparative-HPLC system to obtain the final product **10** (408 mg, yield 88%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.07 (s, 1H), 8.51 (s, 1H), 7.45 (d, *J* = 1.2 Hz, 1H), 7.42 (t, *J* = 3.5 Hz, 3H), 7.33 (dd, *J* = 8.5, 5.5 Hz, 2H), 7.14 – 7.07 (m, 2H), 6.98 (d, *J* = 7.1 Hz, 1H), 6.90 (d, *J* = 3.2 Hz, 1H), 6.64 (d, *J* = 7.1 Hz, 1H), 5.51 (s, 2H), 4.19 (s, 3H), 3.01 (s, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 163.0 (d, *J* = 243.9 Hz), 157.1, 138.7, 135.2 (d, *J* = 3.2 Hz), 131.3, 130.1, 130.0 (d, *J* = 8.2 Hz), 129.8, 129.5, 125.8, 124.8, 122.3, 119.2, 116.2 (d, *J* = 21.6 Hz), 113.4, 106.5, 100.8, 99.9, 50.0, 39.7, 35.8. HRMS (ESI) calcd for $C_{24}H_{22}FN_4O_3S$ $[M + H]^+$, 465.1391; found, 465.1377.

N-(1-(4-fluorobenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**11**). This compound was obtained by a procedure similar to the preparation of **10** (52 mg, yield 89%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.85 (s, 1H), 8.58

(s, 1H), 7.49 (s, 2H), 7.45 – 7.38 (m, 2H), 7.36 – 7.26 (m, 2H), 7.15 – 7.04 (m, 3H), 6.96 (d, *J* = 3.0 Hz, 1H), 6.77 (d, *J* = 7.0 Hz, 1H), 5.47 (s, 2H), 4.20 (s, 3H), 3.14 (q, *J* = 7.3 Hz, 2H), 1.31 (t, *J* = 7.4 Hz, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 163.0 (d, *J* = 244.1 Hz), 157.0, 138.7, 135.2 (d, *J* = 3.3 Hz), 131.4, 130.9, 130.7, 130.0 (d, *J* = 8.2 Hz), 129.8, 129.4, 125.7, 124.3, 121.9, 119.4, 116.2 (d, *J* = 21.8 Hz), 112.4, 106.2, 102.1, 99.8, 50.0, 46.4, 36.0, 8.7. HRMS (ESI) calcd for $C_{25}H_{24}FN_4O_3S$ [M + H]$^+$, 479.1548; found, 479.1525.

N-(1-(2,6-dimethylbenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**19**). This compound was obtained by a procedure similar to the preparation of **10** (35 mg, yield 94%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.18 (s, 1H), 8.48 (s, 1H), 7.55 – 7.49 (m, 2H), 7.47 (s, 1H), 7.23 (dd, *J* = 8.5, 6.5 Hz, 1H), 7.16 (s, 1H), 7.14 (s, 1H), 7.02 (d, *J* = 7.0 Hz, 1H), 6.83 (d, *J* = 3.2 Hz, 1H), 6.77 (d, *J* = 3.3 Hz, 1H), 6.74 (d, *J* = 7.0 Hz, 1H), 5.44 (s, 2H), 4.22 (s, 3H), 3.13 (q, *J* = 7.4 Hz, 2H), 2.29 (s, 6H), 1.30 (t, *J* = 7.3 Hz, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 155.3, 139.1, 138.9, 133.5, 131.3, 129.9, 129.5, 129.2, 127.3, 127.2, 125.8, 124.9, 121.6, 119.5, 112.5, 112.4, 105.8, 101.1, 99.3, 46.4, 45.2, 35.9, 20.0, 8.7. HRMS (ESI) calcd for $C_{27}H_{29}N_4O_3S$ [M + H]$^+$, 489.1955; found, 489.1954.

N-(1-(2,6-dimethylbenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**20**). This compound was obtained by a procedure similar to the preparation of **10** (22 mg, yield 90%). ^{1}H NMR (400 MHz, DMSO-d_6) δ 10.96 (d, *J* = 5.6 Hz, 1H), 9.58 (s, 1H), 7.56 (s, 1H), 7.43 (s, 1H), 7.25 – 7.17 (m, 2H), 7.12 (d, *J* = 7.5 Hz, 2H), 6.94 – 6.88 (m, 1H), 6.75 (dd, *J* = 28.1, 3.2 Hz, 2H), 6.56 (d, *J* = 7.0 Hz, 1H), 5.36 (s, 2H), 4.13 (s, 3H), 2.97 (s, 3H), 2.22 (s, 6H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 155.9, 137.7, 137.6, 132.6, 130.0, 129.3, 128.5, 128.3, 128.3, 128.1, 126.7, 125.3, 123.4, 120.9, 117.5, 112.2, 105.0, 99.6, 99.3, 44.1, 35.4, 19.5. HRMS (ESI) calcd for $C_{26}H_{27}N_4O_3S$ [M + H]$^+$, 475.1798; found, 475.1792.

N-(1-(4-fluoro-2-methylbenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (21). This compound was obtained by a

procedure similar to the preparation of **10** (43 mg, yield 92%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.51 (s, 1H), 8.54 (s, 1H), 7.47 (d, *J* = 1.3 Hz, 1H), 7.45 (s, 1H), 7.38 (t, *J* = 1.0 Hz, 1H), 7.27 (d, *J* = 3.3 Hz, 1H), 7.08 – 7.00 (m, 2H), 6.92 (dd, *J* = 3.3, 0.9 Hz, 1H), 6.92 – 6.81 (m, 2H), 6.73 (d, *J* = 7.0 Hz, 1H), 5.48 (s, 2H), 4.19 (s, 3H), 3.03 (s, 3H), 2.36 (s, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 163.0 (d, *J* = 243.8 Hz), 157.0, 139.7 (d, *J* = 8.1 Hz), 139.0, 132.8 (d, *J* = 2.9 Hz), 131.3, 130.5, 130.3, 130.0 (d, *J* = 8.7 Hz), 129.9, 129.4, 125.8, 124.5, 122.2, 119.3, 117.8 (d, *J* = 21.6 Hz), 113.5 (d, *J* = 21.2 Hz), 113.3, 106.4, 101.6, 100.0, 48.4, 39.8, 35.9, 19.2. HRMS (ESI) calcd for $C_{25}H_{24}FN_4O_3S$ [M + H]$^+$, 479.1548; found, 479.1515.

N-(1-(3-fluorobenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**22**). This compound was obtained by a procedure similar to the preparation of **10** (45 mg, yield 97%). ^{1}H NMR (400 MHz, Methanol-d_4) δ 10.73 (s, 1H), 8.10 (s, 1H), 7.46 (t, *J* = 1.1 Hz, 1H), 7.31 (dd, *J* = 3.0, 1.6 Hz, 1H), 7.26 (s, 1H), 7.15 (td, *J* = 8.2, 6.1 Hz, 1H), 7.09 (d, *J* = 1.3 Hz, 1H), 7.07 – 6.99 (m, 2H), 6.91 – 6.84 (m, 2H), 6.59 (d, *J* = 7.0 Hz, 1H), 6.56 (dt, *J* = 3.1, 1.2 Hz, 1H), 4.98 (s, 2H), 4.07 (s, 3H), 3.07 (s, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 163.9 (d, *J* = 244.6 Hz), 156.4, 142.1 (d, *J* = 6.8 Hz), 138.8, 132.3, 132, 131.5 (d, *J* = 8.3 Hz), 129.8, 129.4, 125.6, 123.8 (d, *J* = 2.3 Hz), 123.6, 122.4, 119.5, 115.1 (d, *J* = 21.3 Hz), 114.7 (d, *J* = 22.0 Hz), 113.4, 106.6, 103.8, 103.7, 100.1, 50.2, 39.8, 36.2. HRMS (ESI) calcd for $C_{24}H_{22}FN_4O_3S$ [M + H]$^+$, 465.1391; found, 465.1373.

N-(6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1-((3-methyloxetan-3-yl)methyl)-1*H*-indol-4-yl)methanesulfonamide (23). This compound was obtained by a procedure similar to the preparation of 10 (30 mg, yield 92%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.81 (s, br, 1H), 8.49 (s, 1H), 7.68 (s, 1H), 7.55 (s, 1H), 7.46 (d, *J* = 1.1 Hz, 1H), 7.37 (d, *J* = 3.3 Hz, 1H), 7.10 (d, *J* = 7.0 Hz, 1H), 7.01 (d, *J* = 7.0 Hz, 1H), 6.90 (d, *J* = 2.8 Hz, 1H), 4.39 – 4.28 (m, 2H), 4.22 (s, 3H), 3.52 – 3.46 (m, 3H), 3.40 (d, *J* = 10.0 Hz, 1H), 3.01 (s, 3H), 1.03 (s, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 157.4, 140.2, 131.04, 130.98, 130.8, 130.5, 130.4, 129.7,

125.7, 121.8, 119.5, 113.3, 106.8, 102.3, 99.9, 65.9, 50.0, 42.0, 39.8, 36.0, 21.7. HRMS (ESI) calcd for $C_{22}H_{25}N_4O_4S$ [M + H]$^+$, 441.1591; found, 441.1632.

N-(1-((3,3-difluorocyclobutyl)methyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (24). This compound was obtained by a procedure similar to the preparation of **10** (41 mg, yield 89%). ^{1}H NMR (400 MHz, DMSO-d_6) δ 10.97 (d, *J* = 5.6 Hz, 1H), 9.61 (d, *J* = 1.7 Hz, 1H), 7.59 (s, 1H), 7.44 (s, 1H), 7.40 (d, *J* = 3.2 Hz, 1H), 7.24 (d, *J* = 1.2 Hz, 1H), 6.93 (dd, *J* = 7.0, 5.7 Hz, 1H), 6.76 (d, *J* = 3.2 Hz, 1H), 6.71 (d, *J* = 6.9 Hz, 1H), 4.32 (d, *J* = 6.7 Hz, 2H), 4.13 (s, 3H), 2.98 (s, 3H), 2.74 – 2.55 (m, 3H), 2.48 – 2.35 (m, 2H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 155.9, 137.4, 130.0, 129.4, 128.4, 128.3, 125.4, 123.3, 120.8, 120.4 (dd, *J* = 283.8, 273.4 Hz), 117.5, 112.0, 104.8, 99.6, 99.3, 49.5, 38.2 (t, *J* = 22.0 Hz), 35.4, 23.2 (dd, *J* = 12.6, 5.9 Hz). HRMS (ESI) calcd for $C_{22}H_{23}F_2N_4O_3S$ [M + H]$^+$, 461.1453; found, 461.1462.

7-bromo-1-(4-fluorobenzyl)-5-nitro-1*H*-indole (**42a**). To a solution of **41** (1.21 g, 5 mmol) in DMF (20 mL) at RT was added K_2CO_3 (968 mg, 7 mmol) and **52** (0.75 mL, 6 mmol). The mixture was then stirred at RT for 2 h, followed by quenching in water. The resulting solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 4:1) to obtain **42a** (1.53 g, yield 88%) as a pale yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.53 (d, *J* = 2.1 Hz, 1H), 8.28 (d, *J* = 2.1 Hz, 1H), 7.24 (d, *J* = 3.3 Hz, 1H), 7.02 – 6.96 (m, 4H), 6.78 (d, *J* = 3.3 Hz, 1H), 5.83 (s, 2H). LCMS (m/z) [M + H]$^+$ = 349.1.

7-bromo-1-(4-fluoro-2-methylbenzyl)-5-nitro-1*H*-indole (**42b**). This compound was obtained by a procedure similar to the preparation of **42a** (430 mg, yield 96%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.54 (t, *J* = 2.2 Hz, 1H), 8.28 (d, *J* = 2.7 Hz, 1H), 7.14 (d, *J* = 3.2 Hz, 1H), 6.95

(d, *J* = 9.5 Hz, 1H), 6.77 (q, *J* = 3.4, 2.7 Hz, 2H), 6.40 (d, *J* = 6.9 Hz, 1H), 5.76 (s, 2H), 2.33 (s, 3H). LCMS (m/z) $[M + H]^+$ = 363.1.

7-bromo-1-(4-fluorobenzyl)-1*H*-indol-5-amine (**43a**). To a solution of **42a** (1.04 g, 3 mmol) in EtOH (15 mL) at RT was added iron powder (1.34 g, 24 mmol) and saturated ammonia chloride (3 mL). The reaction was refluxed for 2 h and then cooled to room temperature. The mixture was filtered to obtain a clear solution, followed by extraction with ethyl acetate. Organic layers were combined and dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave **43a** as a brown solid without further purification (940 mg). LCMS (m/z) $[M + H]^+$ = 319.1.

7-bromo-1-(4-fluoro-2-methylbenzyl)-1*H*-indol-5-amine (**43b**). This compound was obtained by a procedure similar to the preparation of **43a** (160 mg, yield 66%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 6.94 – 6.87 (m, 3H), 6.83 (d, *J* = 2.4 Hz, 1H), 6.72 (t, *J* = 8.7 Hz, 1H), 6.35 (q, *J* = 5.9, 4.5 Hz, 2H), 5.60 (s, 2H), 2.32 (d, *J* = 2.0 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 333.1.

7-bromo-1-(4-fluoro-2,6-dimethylbenzyl)-1*H*-indol-5-amine (**43c**). This compound was obtained by a procedure similar to the preparation of **43a** (129 mg, yield 90%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.49 (d, *J* = 2.2 Hz, 1H), 8.35 (d, *J* = 2.2 Hz, 1H), 6.87 (d, *J* = 9.2 Hz, 2H), 6.67 (d, *J* = 3.4 Hz, 1H), 6.58 (d, *J* = 3.4 Hz, 1H), 5.87 (s, 2H), 2.28 (s, 6H). LCMS (m/z) $[M + H]^+$ = 347.1.

N-(7-bromo-1-(4-fluorobenzyl)-1*H*-indol-5-yl)methanesulfonamide (**44a**). To a solution of **43a** (957 mg, 3 mmol) in pyridine (10 mL) at 0 °C was added MsCl (0.675 mL, 9 mmol). The reaction mixture was allowed to warm to RT and stirred for 3 h. Water was then added to quench the reaction. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 1:1) to obtain **44a**

(1.06 g, yield 89%) as a grey solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.53 (d, *J* = 2.0 Hz, 1H), 7.27 (s, 2H), 7.15 (d, *J* = 3.2 Hz, 1H), 6.99 (s, 1H), 6.97 (s, 2H), 6.56 (d, *J* = 3.2 Hz, 1H), 5.76 (d, *J* = 1.1 Hz, 2H), 3.00 (s, 3H). LCMS (m/z) $[M + H]^+$ = 396.9.

N-(7-bromo-1-(4-fluoro-2-methylbenzyl)-1*H*-indol-5-yl)methanesulfonamide (**44b**). This compound was obtained by a procedure similar to the preparation of **44a** (118 mg, yield 83%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.55 (d, *J* = 2.0 Hz, 1H), 7.27 (s, 1H), 7.04 (d, *J* = 3.3 Hz, 1H), 6.94 (dd, *J* = 9.5, 2.7 Hz, 1H), 6.75 (td, *J* = 8.4, 2.8 Hz, 1H), 6.55 (d, *J* = 3.2 Hz, 1H), 6.47 (s, 1H), 6.37 (dd, *J* = 8.6, 5.7 Hz, 1H), 5.69 (s, 2H), 3.01 (s, 3H), 2.34 (s, 3H). LCMS (m/z) $[M + H]^+$ = 411.1.

N-(7-bromo-1-(4-fluoro-2,6-dimethylbenzyl)-1*H*-indol-5-yl)methanesulfonamide (**44c**). This compound was obtained by a procedure similar to the preparation of **44a** (103 mg, yield 90%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.58 (d, *J* = 2.2 Hz, 1H), 8.28 (d, *J* = 2.2 Hz, 1H), 6.99 (d, *J* = 9.5 Hz, 2H), 6.86 (d, *J* = 3.4 Hz, 1H), 6.78 (d, *J* = 3.4 Hz, 1H), 6.00 (s, 2H), 2.80 – 2.76 (m, 4H), 2.32 (d, *J* = 0.8 Hz, 6H). LCMS (m/z) $[M + H]^+$ = 424.1.

N-(1-(4-fluorobenzyl)-7-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (**45a**). A mixture of **44a** (792 mg, 2 mmol), Pd(dppf)Cl_2-DCM (82 mg, 0.1 mmol), **35** (694 mg, 2.4 mmol), and K_3PO_4 (1.1 g, 5.2 mmol) in dioxane-water (15 mL, V/V, 4:1) was heated at 90 °C for 24 h. Upon completion of the reaction, the reaction mixture was then diluted with ethyl acetate and filtered to obtain a clear solution, which was subsequently washed three times with brine. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by flash column chromatography on silica gel (Hexanes-Acetone, 1:1) to yield **45a** (832 mg, yield 87%) as a grey solid. ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.24 (s, 1H), 7.63 (d, *J* = 2.1 Hz, 1H), 7.58 (d, *J* = 5.7 Hz, 1H), 7.38 (d, *J* = 3.2 Hz, 1H), 7.01 (d, *J* = 2.2 Hz, 1H), 6.94 (s, 1H), 6.83 – 6.78 (m, 2H), 6.69 (d, *J* = 5.6 Hz, 1H), 6.63 (d,

J = 3.2 Hz, 1H), 6.43 – 6.38 (m, 2H), 4.07 (s, 3H), 4.05 (s, 3H), 2.93 (d, *J* = 1.1 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 479.2.

N-(1-(4-fluoro-2-methylbenzyl)-7-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (**45b**). This compound was obtained by a procedure similar to the preparation of **45a** (123 mg, yield 91%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.25 (s, 1H), 7.67 (d, *J* = 2.2 Hz, 1H), 7.57 (d, *J* = 5.6 Hz, 1H), 7.28 (d, *J* = 3.2 Hz, 1H), 7.00 (d, *J* = 2.2 Hz, 1H), 6.78 (dd, *J* = 9.8, 2.7 Hz, 1H), 6.72 – 6.66 (m, 3H), 6.65 (d, *J* = 3.2 Hz, 1H), 6.04 (dd, *J* = 8.6, 5.9 Hz, 1H), 4.91 (s, 2H), 4.06 (s, 3H), 3.93 (s, 3H), 2.94 (s, 3H), 1.62 (s, 3H). LCMS (m/z) $[M + H]^+$ = 493.1.

N-(1-(4-fluoro-2,6-dimethylbenzyl)-7-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (45c). This compound was obtained by a procedure similar to the preparation of **45a** (104 mg, yield 89%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.27 (s, 1H), 7.69 (d, *J* = 5.7 Hz, 1H), 7.60 (d, *J* = 2.2 Hz, 1H), 7.55 (s, 1H), 7.12 (d, *J* = 2.2 Hz, 1H), 6.95 (d, *J* = 5.6 Hz, 1H), 6.81 (d, *J* = 9.6 Hz, 2H), 6.52 (d, *J* = 3.3 Hz, 1H), 6.45 (d, *J* = 3.3 Hz, 1H), 4.18 (s, 3H), 4.06 (s, 3H), 2.95 (s, 3H), 2.02 (s, 6H). LCMS (m/z) $[M + H]^+$ = 507.2.

N-(1-(4-fluorobenzyl)-7-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (**12**). A mixture of **45a** (478 mg, 1 mmol), TMSCl (0.2 mL, 1.6 mmol), NaI (240 mg, 1.6 mmol), and H_2O (9 μL, 0.5 mmol) in acetonitrile (8 mL) was heated at 50 °C for 2 h. The reaction was then cooled to room temperature. The reaction mixture was diluted with ethyl acetate and washed three times with aqueous $NaHCO_3$. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by preparative-LC system to obtain the final product **12** (417 mg) with a 90% yield. ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.18 (s, 1H), 8.30 (s, 1H), 7.62 (d, *J* = 2.1 Hz, 1H), 7.40 (d, *J* = 3.2 Hz, 1H), 6.98 (d, *J* = 2.2 Hz, 1H), 6.93 – 6.87 (m, 2H), 6.84 (t, *J* = 8.8 Hz, 2H), 6.63 (d, *J* = 3.2 Hz, 1H), 6.45 (dd, *J* = 8.4, 5.4 Hz, 2H), 6.00 (d, *J* = 6.8 Hz, 1H), 5.10 (s, 2H), 4.11 (s, 3H), 2.92 (s, 3H). ^{13}C

NMR (101 MHz, Acetone-d_6) δ 162.5 (d, J = 242.9 Hz), 143.3, 136.1 (d, J = 2.1 Hz), 133.2, 133.0, 132.0, 131.7, 131.2, 127.9 (d, J = 8.1 Hz), 126.0, 123.4, 122.2, 120.0, 115.6 (d, J = 21.7 Hz), 115.0, 114.7, 102.7, 100.7, 51.6, 38.8, 35.8. HRMS (ESI) calcd for $C_{24}H_{22}FN_4O_3S$ $[M + H]^+$, 465.1391; found, 465.1380.

N-(1-(4-fluoro-2-methylbenzyl)-7-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (13). This compound was obtained by a procedure similar to the preparation of **12** (55 mg, yield 95%). ^{1}H NMR (400 MHz, DMSO-d_6) δ 10.84 (d, J = 5.6 Hz, 1H), 9.38 (s, 1H), 7.49 (d, J = 2.1 Hz, 1H), 7.38 (d, J = 3.1 Hz, 1H), 6.85 (dd, J = 10.0, 2.7 Hz, 1H), 6.75 (d, J = 2.1 Hz, 1H), 6.74 – 6.66 (m, 3H), 6.61 (d, J = 3.1 Hz, 1H), 5.86 – 5.78 (m, 2H), 4.91 (s, 2H), 3.90 (s, 3H), 2.92 (s, 3H), 1.70 (s, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 160.7 (d, J = 241.7 Hz), 155.6, 136.1 (d, J = 7.9 Hz), 133.3 (d, J = 2.0 Hz), 132.3, 131.7, 131.2, 130.4, 130.1, 130.0, 125.8 (d, J = 8.7 Hz), 125.3, 121.8, 120.6, 118.5, 115.8 (d, J = 21.0 Hz), 113.0, 112.8, 112.2 (d, J = 21.0 Hz), 101.5, 99.0, 48.4, 38.6, 35.0, 17.5. HRMS (ESI) calcd for $C_{25}H_{24}FN_4O_3S$ $[M + H]^+$, 479.1548; found, 479.1560.

N-(1-(4-fluoro-2,6-dimethylbenzyl)-7-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-5-yl)methanesulfonamide (14). This compound was obtained by a procedure similar to the preparation of **12** (47 mg, yield 92%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.03 (s, 1H), 8.30 (s, 1H), 7.59 (d, J = 2.2 Hz, 1H), 7.46 (s, 1H), 7.10 (d, J = 2.2 Hz, 1H), 7.03 – 6.98 (m, 1H), 6.82 (d, J = 9.6 Hz, 2H), 6.52 (d, J = 3.3 Hz, 1H), 6.44 (d, J = 3.3 Hz, 1H), 6.25 (d, J = 7.0 Hz, 1H), 4.88 (s, 2H), 4.23 (s, 3H), 2.95 (s, 3H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 162.9 (d, J = 244.3 Hz), 157.0, 141.6 (d, J = 8.5 Hz), 134.4, 132.7, 131.5, 131.3, 131.0, 130.1 (d, J = 2.5 Hz), 128.4, 126.3, 123.9, 122.3, 119.7, 115.8, 115.5 (d, J = 21.0 Hz), 114.8, 102.4, 100.5, 46.3, 38.8, 35.9, 19.7. HRMS (ESI) calcd for $C_{26}H_{25}FN_4O_3S$ $[M + H]^+$, 493.1704; found, 493.1686.

6-bromo-1-(4-fluorobenzyl)-4-nitro-1*H*-indazole (**47**). This compound was obtained by a procedure similar to the preparation of **42a** (301 mg, yield 83%). ^{1}H NMR (400 MHz, Chloroform-

d) δ 8.62 (d, *J* = 1.0 Hz, 1H), 8.22 (d, *J* = 1.4 Hz, 1H), 7.84 (t, *J* = 1.1 Hz, 1H), 7.24 – 7.17 (m, 2H), 7.08 – 6.99 (m, 2H), 5.60 (s, 2H). LCMS (m/z) $[M + H]^+$ = 349.9.

N-(6-bromo-1-(4-fluorobenzyl)-1*H*-indazol-4-yl)methanesulfonamide (**49a**). This compound was obtained by a procedure similar to the preparation of **44a** (55 mg, yield 95%). LCMS (m/z) $[M + H]^+$ = 397.9

N-(6-bromo-1-(4-fluorobenzyl)-1*H*-indazol-4-yl)ethanesulfonamide (**49b**). This compound was obtained by a procedure similar to the preparation of **44a** (166 mg, yield 86%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.22 (d, *J* = 1.0 Hz, 1H), 7.87 (s, 1H), 7.34 (d, *J* = 1.3 Hz, 1H), 7.30 (t, *J* = 1.2 Hz, 1H), 7.21 – 7.15 (m, 2H), 7.03 – 6.96 (m, 2H), 5.48 (s, 2H), 3.26 (q, *J* = 7.4 Hz, 2H), 1.39 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 412.0.

N-(1-(4-fluorobenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indazol-4-yl)methanesulfonamide (**50a**). This compound was obtained by a procedure similar to the preparation of **45a** (98 mg, yield 90%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.19 – 8.14 (m, 1H), 7.80 (d, *J* = 5.8 Hz, 1H), 7.46 (d, *J* = 1.1 Hz, 1H), 7.33 (s, 1H), 7.30 (s, 1H), 7.28 – 7.24 (m, 2H), 7.23 (d, *J* = 5.8 Hz, 1H), 7.04 (t, *J* = 8.6 Hz, 2H), 5.60 (s, 2H), 4.13 (s, 6H), 3.15 (s, 3H). LCMS (m/z) $[M + H]^+$ = 480.1.

N-(1-(4-fluorobenzyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indazol-4-yl)ethanesulfonamide (**50b**). This compound was obtained by a procedure similar to the preparation of **45a** (87 mg, yield 77%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.20 (d, *J* = 0.9 Hz, 1H), 7.77 (d, *J* = 5.8 Hz, 1H), 7.57 (s, 1H), 7.46 (d, *J* = 1.1 Hz, 1H), 7.27 (d, *J* = 1.5 Hz, 2H), 7.25 – 7.20 (m, 3H), 7.01 (t, *J* = 8.6 Hz, 2H), 5.57 (s, 2H), 4.10 (d, *J* = 2.0 Hz, 6H), 3.25 (q, *J* = 7.4 Hz, 2H), 1.41 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 494.2.

N-(1-(4-fluorobenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indazol-4-yl)methanesulfonamide (**17**). This compound was obtained by a procedure

similar to the preparation of **12** (38 mg, yield 91%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.05 (s, 1H), 8.94 (s, 1H), 8.34 (d, *J* = 0.9 Hz, 1H), 7.54 (s, 1H), 7.53 (s, 1H), 7.49 (d, *J* = 1.1 Hz, 1H), 7.45 – 7.36 (m, 2H), 7.13 – 7.07 (m, 2H), 7.02 (d, *J* = 7.1 Hz, 1H), 6.70 (d, *J* = 7.1 Hz, 1H), 5.71 (s, 2H), 4.20 (s, 3H), 3.11 (s, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.6 (d, *J* = 243.4 Hz), 155.9, 141.1, 133.8, 133.7 (d, *J* = 3.4 Hz), 131.9, 131.2, 130.1, 129.6 (d, *J* = 8.4 Hz), 128.5, 125.8, 123.6, 116.4, 116.0, 115.4 (d, *J* = 21.5 Hz), 110.3, 102.7, 99.4, 51.2, 39.6, 35.5. HRMS (ESI) calcd for $C_{23}H_{21}FN_5O_3S$ [M + H]$^+$, 466.1344; found, 466.1353.

N-(1-(4-fluorobenzyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indazol-4-yl)ethanesulfonamide (**18**). This compound was obtained by a procedure similar to the preparation of **12** (31 mg, yield 90%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.21 (s, 1H), 9.00 (s, 1H), 8.38 (s, 1H), 7.60 – 7.48 (m, 3H), 7.44 – 7.35 (m, 2H), 7.10 (t, *J* = 8.8 Hz, 2H), 7.05 (d, *J* = 7.1 Hz, 1H), 6.73 (d, *J* = 7.0 Hz, 1H), 5.70 (s, 2H), 4.20 (s, 3H), 3.23 (q, *J* = 7.4 Hz, 2H), 1.32 (t, *J* = 7.4 Hz, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.6 (d, *J* = 243.4 Hz), 155.9, 141.1, 133.8, 133.7 (d, *J* = 3.2 Hz), 131.8, 131.2, 130.1, 129.7 (d, *J* = 8.3 Hz), 128.5, 125.8, 123.6, 116.4, 115.7, 115.4 (d, *J* = 21.4 Hz), 109.8, 102.6, 99.3, 51.1, 45.5, 35.5, 8.2. HRMS (ESI) calcd for $C_{24}H_{23}FN_5O_3S$ [M + H]$^+$, 480.1500; found, 480.1487.

7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridine-3-carbaldehyde (**36**). To a solution of **34** (288 mg, 1 mmol) in THF (6 mL) at -78 °C was added *n*-BuLi (1.2 mmol) for 1 h. The reaction was cooled to -78 °C and DMF (0.1 mL) was added. Dry ice bath was removed, and the mixture was stirred for 2 h at room temperature. Water was added to quench the reaction. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 4:1) to obtain **36** (138 mg, yield 73%) as a white solid. ^{1}H NMR (400 MHz, Acetone-d_6) δ 9.97 (s, 1H), 8.12 – 8.03 (m, 1H), 7.83 (dd, *J* =

5.5, 1.4 Hz, 1H), 7.66 (dt, *J* = 5.8, 1.5 Hz, 1H), 4.14 (s, 3H), 4.05 (s, 3H). LCMS (m/z) $[M + H]^+$ = 191.2.

N[1]-(4-fluorobenzyl)-3-nitrobenzene-1,2-diamine (**53**). To a solution of **51** (153 mg, 1 mmol) in DMF (5 mL) at RT was added K_2CO_3 (276 mg, 2 mmol) and **52** (137 μL, 1.1 mmol). The solution was stirred at RT. After 2 h, the reaction was quenched by addition of water. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 2:1) to obtain **53** (196 mg, yield 75%) as a yellow solid. LCMS (m/z) $[M + H]^+$ = 262.2.

1-(4-fluorobenzyl)-2-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-4-nitro-1*H*-benzo[d]imidazole (**54**). To a solution of **53** (261 mg, 1 mmol) in HOAc-H_2O (6 mL, V/V, 2:1) at RT was added $NaBO_3{\cdot}4H_2O$ (462 mg, 3 mmol). The solution was stirred at 50 °C for 1 h and then quenched by addition of water. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-acetone, 1:1) to obtain **54** (289 mg, yield 67%) as a yellow solid. ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.15 – 8.04 (m, 3H), 7.96 (d, *J* = 8.1 Hz, 1H), 7.85 (d, *J* = 5.6 Hz, 1H), 7.38 (t, *J* = 8.0 Hz, 1H), 7.13 (d, *J* = 7.1 Hz, 4H), 5.86 (s, 2H), 4.07 (s, 3H), 4.03 (s, 3H). LCMS (m/z) $[M + H]^+$ = 432.1.

1-(4-fluorobenzyl)-2-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-benzo[*d*]imidazol-4-amine (**55**). To a solution of **54** (1.29 g, 3 mmol) in EtOH (15 mL) at RT was added iron powder (1.34 g, 24 mmol) and saturated ammonia chloride (3 mL). The reaction was refluxed for 2 h and then cooled to room temperature. The mixture was filtered to obtain a clear solution, followed by extraction with ethyl acetate. Organic layers were combined and dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave **55** as a brown solid without further purification (905 mg). LCMS (m/z) $[M + H]^+$ = 402.1.

N-(1-benzyl-2-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-benzo[*d*]imidazol-4-yl)ethanesulfonamide (56a). **N-(1-benzyl-2-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-benzo[*d*]imidazol-4-yl)-N-(ethylsulfonyl)ethanesulfonamide (56b)**. To a solution of **55** (431 mg, 1 mmol) in DCM (5 mL) at 0 °C was added $EtSO_2Cl$ (0.28 mL, 3 mmol). Then ice bath was removed. After stirring for 3 h at RT, the reaction contents were quenched by the addition of water. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue without further purification. For **56a**, LCMS (m/z) $[M + H]^+$ = 494.2, For **56b**, LCMS (m/z) $[M + H]^+$ = 586.1.

N-(1-(4-fluorobenzyl)-2-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-benzo[*d*]imidazol-4-yl)ethanesulfonamide (15). This compound was obtained by a procedure similar to the preparation of **10** (26 mg, yield 37% for two steps). ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.11 (d, *J* = 5.9 Hz, 1H), 9.60 (s, 1H), 7.72 (s, 1H), 7.31 – 7.27 (m, 2H), 7.17 – 7.09 (m, 6H), 6.99 (dd, *J* = 7.0, 5.7 Hz, 1H), 5.68 (s, 2H),3.37 (q, *J* = 7.4 Hz, 2H), 1.34 (t, *J* = 7.3 Hz, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.4 (d, *J* = 243.4 Hz), 155.6, 147.8, 136.8, 136.6, 136.4, 133.1 (d, *J* = 2.5 Hz), 131.3, 131.0, 128.4 (d, *J* = 8.3 Hz), 128.1, 126.1, 123.4, 122.4, 115.7 (d, *J* = 21.9 Hz), 107.2, 105.0, 101.3, 46.7, 46.6, 35.8, 8.3. HRMS (ESI) calcd for $C_{24}H_{23}FN_5O_3S$ $[M + H]^+$, 480.1500; found, 480.1496.

N-(ethylsulfonyl)-N-(1-(4-fluorobenzyl)-2-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-benzo[*d*]imidazol-4-yl)ethanesulfonamide (16). This compound was obtained by a procedure similar to the preparation of **10** (32 mg, yield 31% for two steps). ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.15 (d, *J* = 5.6 Hz, 1H), 7.79 (s, 1H), 7.64 – 7.59 (m, 1H), 7.34 (dd, *J* = 7.8, 1.0 Hz, 1H), 7.26 (t, *J* = 7.9 Hz, 1H), 7.19 – 7.11 (m, 5H), 7.02 (dd, *J* = 6.9, 5.6 Hz, 1H), 5.74 (s, 2H), 4.10 (s, 3H), 3.95 – 3.78 (m, *J* = 7.2 Hz, 4H), 1.38 (t, *J* = 7.4 Hz, 6H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.5 (d, *J* = 243.5 Hz), 155.6, 149.6, 142.0, 136.8, 132.8 (d, *J*

= 2.1 Hz), 131.5, 131.4, 128.5 (d, *J* = 8.2 Hz), 126.6, 125.4, 123.6, 123.5, 122.2, 115.8 (d, *J* = 21.6 Hz), 112.5, 104.6, 100.7, 50.5, 46.8, 35.9, 7.7. HRMS (ESI) calcd for $C_{26}H_{27}FN_5O_5S_2$ [M + H]$^+$, 572.1432; found, 572.1424.

6-bromo-1-(di(pyridin-2-yl)methyl)-4-nitro-1*H*-indole (**58a**). To a solution of **57** (7.23 g, 30 mmol) in DMF (60 mL) at RT was added K_2CO_3 (6.22 g, 45 mmol) and 2,2'-(bromomethylene)dipyridine (8.97 g, 36 mmol)[83] and then the mixture was stirred at 50 °C for 6 h. Water was added to quench the reaction. The solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 2:1) to obtain **58a** (11.17 g, yield 91%) as a yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.70 – 8.64 (m, 2H), 8.25 (d, *J* = 1.5 Hz, 1H), 7.85 (d, *J* = 1.4 Hz, 1H), 7.72 (td, *J* = 7.8, 1.8 Hz, 2H), 7.52 (d, *J* = 3.4 Hz, 1H), 7.33 – 7.25 (m, 3H), 7.16 (d, *J* = 7.9 Hz, 2H), 6.97 (s, 1H). LCMS (m/z) [M + H]$^+$ = 409.0.

1-benzhydryl-6-bromo-4-nitro-1*H*-indole (**58b**). This compound was obtained by a procedure similar to the preparation of **58a** (425 mg, yield 72%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.26 (d, *J* = 1.5 Hz, 1H), 7.69 (t, *J* = 1.1 Hz, 1H), 7.40 – 7.34 (m, 6H), 7.21 (dd, *J* = 3.3, 0.8 Hz, 1H), 7.11 – 7.05 (m, 5H), 6.82 (s, 1H). LCMS (m/z) [M + H]$^+$ = 407.0.

6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-4-nitro-1*H*-indole (**59a**). To a solution of **58a** (10.23 g, 25 mmol) in THF (80 mL) at 0 °C was added *t*-BuOK (5.61 g, 50 mmol) and MeI (4.67 mL, 75 mmol) successively. Then ice bath was removed. The solution was stirred at RT for 2 h and water was added to quench the reaction. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 3:1) to obtain **59a** (9.84 g, yield 93%) as a yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.66 (d, *J* = 4.7 Hz, 2H), 8.15 (d, *J* = 1.5 Hz, 1H), 7.66 (td, *J* = 7.8, 1.9 Hz, 2H), 7.28 (t, *J* =

3.4 Hz, 3H), 7.26 (s, 1H), 7.11 (d, *J* = 8.1 Hz, 2H), 6.91 (s, 1H), 2.55 (s, 3H). LCMS (m/z) $[M + H]^+$ = 423.1.

6-bromo-1-(1,1-di(pyridin-2-yl)propyl)-4-nitro-1*H*-indole (**59b**). This compound was obtained by a procedure similar to the preparation of **59a** (276 mg, yield 73%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.65 – 8.59 (m, 2H), 8.15 (s, 1H), 7.66 (td, *J* = 7.8, 1.9 Hz, 2H), 7.51 – 7.43 (m, 3H), 7.28 – 7.18 (m, 3H), 6.82 (s, 1H), 3.10 (q, *J* = 7.3 Hz, 2H), 0.70 (t, *J* = 7.3 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 437.1.

6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indol-4-amine (**60a**). To a solution of **59a** (9.30 g, 22 mmol) in EtOH (60 mL) at RT was added iron powder (9.86 g, 176 mmol) and saturated ammonia chloride (15 mL). The reaction was refluxed for 2 h, and then cooled to room temperature. The mixture was filtered to obtain a clear solution, followed by extraction with ethyl acetate. Organic layers were combined and dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave **60a** (7.93 g) as a brown solid without further purification. LCMS (m/z) $[M + H]^+$ = 393.1.

N-(6-bromo-1-(1,1-diphenylethyl)-7,7a-dihydro-1*H*-indol-4-yl)ethanesulfonamide (**61a**). This compound was obtained by a procedure similar to the preparation of **61c** (202 mg, yield 83%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.42 – 7.40 (m, 2H), 7.34 (dd, *J* = 5.2, 1.9 Hz, 6H), 7.25 (t, *J* = 1.1 Hz, 1H), 7.11 – 7.05 (m, 4H), 6.84 (d, *J* = 3.4 Hz, 1H), 6.74 (s, 1H), 6.65 (d, *J* = 3.4 Hz, 1H), 3.20 (q, *J* = 7.4 Hz, 2H), 1.37 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 469.1.

N-(6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indol-4-yl)methanesulfonamide (**61b**). This compound was obtained by a procedure similar to the preparation of **61c** (299 mg, yield 90%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.66 (ddd, *J* = 4.8, 1.9, 0.9 Hz, 2H), 7.63 (td, *J* = 7.8, 1.9 Hz, 2H), 7.29 (d, *J* = 1.4 Hz, 1H), 7.24 (dd, *J* = 2.8, 1.1 Hz, 1H), 7.23 (d, *J* = 1.1 Hz, 1H), 7.16

(s, 1H), 7.07 (s, 1H), 7.05 (s, 1H), 6.98 (d, *J* = 3.5 Hz, 1H), 6.57 (dd, *J* = 3.6, 0.9 Hz, 1H), 6.46 (t, *J* = 1.2 Hz, 1H), 3.04 (s, 3H), 2.52 (d, *J* = 1.8 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 471.1.

N-(6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indol-4-yl)ethanesulfonamide (**61c**). To a solution of **60a** (from previous step) in pyridine (40 mL) at 0 °C was added $EtSO_2Cl$ (4.17 mL, 44 mmol). Upon completion monitored by TLC, the reaction contents were quenched by the addition of water. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 1:1) to obtain **61c** (9.18 g, yield 86% for two steps) as a dark yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.68 (ddd, *J* = 4.8, 2.0, 1.0 Hz, 2H), 7.65 (tt, *J* = 7.8, 1.6 Hz, 2H), 7.33 (d, *J* = 1.3 Hz, 1H), 7.27 (ddt, *J* = 7.4, 4.8, 1.2 Hz, 2H), 7.07 (d, *J* = 7.9 Hz, 2H), 6.98 (dd, *J* = 3.5, 1.3 Hz, 1H), 6.64 (ddd, *J* = 7.1, 3.6, 0.9 Hz, 1H), 6.46 (t, *J* = 1.1 Hz, 1H), 3.20 (q, *J* = 7.4 Hz, 2H), 2.54 (s, 3H), 1.39 (td, *J* = 7.4, 1.2 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 485.2.

N-(6-bromo-1-(1,1-di(pyridin-2-yl)propyl)-7,7a-dihydro-1*H*-indol-4-yl)ethanesulfonamide (**61d**). This compound was obtained by a procedure similar to the preparation of **61c** (183 mg, yield 85%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.60 (dd, *J* = 5.1, 1.8 Hz, 2H), 7.61 (td, *J* = 7.8, 1.9 Hz, 2H), 7.47 (s, 1H), 7.43 (dd, *J* = 8.0, 1.3 Hz, 2H), 7.31 (d, *J* = 1.4 Hz, 1H), 7.20 – 7.15 (m, 3H), 6.66 (d, *J* = 3.5 Hz, 1H), 6.34 (s, 1H), 3.18 (q, *J* = 7.4 Hz, 2H), 3.08 (q, *J* = 7.2 Hz, 2H), 1.35 (t, *J* = 7.4 Hz, 3H), 0.66 (t, *J* = 7.3 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 501.1.

N-(6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indol-4-yl)isobutyramide (**61e**). This compound was obtained by a procedure similar to the preparation of **61c** (138 mg, yield 88%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.64 (dd, *J* = 4.7, 1.8 Hz, 2H), 7.95 (s, 1H), 7.59 (td, *J* = 7.8, 1.9 Hz, 2H), 7.55 (s, 1H), 7.21 (dd, *J* = 7.6, 4.8 Hz, 2H), 7.02 (d, *J* = 8.0 Hz, 2H), 6.94 (d, *J* = 3.5 Hz,

1H), 6.42 (d, *J* = 3.6 Hz, 1H), 6.39 (s, 1H), 2.60 (p, *J* = 6.9 Hz, 1H), 2.50 (s, 3H), 1.25 (d, *J* = 6.9 Hz, 6H). LCMS (m/z) $[M + H]^+$ = 463.1.

N-(6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indol-4-yl)acetamide (**61f**). This compound was obtained by a procedure similar to the preparation of **61c** (106 mg, yield 86%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 9.06 (s, 1H), 8.62 (ddd, *J* = 4.8, 1.9, 1.0 Hz, 2H), 8.10 (s, 1H), 7.76 (td, *J* = 7.8, 1.9 Hz, 2H), 7.34 (ddd, *J* = 7.5, 4.8, 1.1 Hz, 2H), 7.15 (dt, *J* = 8.1, 1.1 Hz, 2H), 7.02 (d, *J* = 3.6 Hz, 1H), 6.81 – 6.72 (m, 1H), 6.33 (t, *J* = 1.3 Hz, 1H), 2.50 (s, 3H), 2.17 (s, 3H). LCMS (m/z) $[M + H]^+$ = 435.0.

N-(1-benzhydryl-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**62a**). This compound was obtained by a procedure similar to the preparation of **62c** (75 mg, yield 66%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.66 (d, *J* = 5.8 Hz, 1H), 7.48 (d, *J* = 1.2 Hz, 1H), 7.35 (qd, *J* = 4.0, 2.3 Hz, 7H), 7.25 (d, *J* = 1.1 Hz, 1H), 7.18 – 7.15 (m, 5H), 6.96 (d, *J* = 5.8 Hz, 1H), 6.84 (d, *J* = 3.4 Hz, 1H), 6.81 (s, 1H), 6.64 (d, *J* = 3.1 Hz, 1H), 4.07 (s, 3H), 4.03 (s, 3H), 3.20 (q, *J* = 7.4 Hz, 2H), 1.38 (t, *J* = 7.4 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 551.2.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (**62b**). This compound was obtained by a procedure similar to the preparation of **62c** (109 mg, yield 91%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.66 (ddd, *J* = 4.7, 1.9, 0.9 Hz, 2H), 8.45 (s, 1H), 7.78 (td, *J* = 7.8, 1.9 Hz, 2H), 7.57 (d, *J* = 5.7 Hz, 1H), 7.40 (d, *J* = 1.2 Hz, 1H), 7.37 (ddd, *J* = 7.6, 4.8, 1.1 Hz, 2H), 7.30 (d, *J* = 1.1 Hz, 1H), 7.27 (s, 2H), 7.07 (d, *J* = 3.5 Hz, 1H), 6.89 (d, *J* = 3.5 Hz, 1H), 6.71 (d, *J* = 5.8 Hz, 1H), 6.66 (t, *J* = 1.1 Hz, 1H), 4.05 (s, 3H), 4.01 (s, 3H), 3.03 (s, 3H), 2.55 (s, 3H). LCMS (m/z) $[M + H]^+$ = 553.2.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**62c**). A mixture of **61c** (7.28 g, 15 mmol), Pd(dppf)Cl_2-

DCM (612 mg, 0.75 mmol), **35** (5.20 g, 18 mmol), and K_3PO_4 (8.28 g, 39 mmol) in dioxane-water (80 mL, 4:1) was heated at 90 °C for 24 h. The mixture was then diluted with ethyl acetate and filtered to obtain a clear solution, which was washed three times with brine. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by flash column chromatography on silica gel (Hexanes-Acetone, 1:1) to yield **62c** (7.48 g, yield 88%) as a grey solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.69 (ddd, *J* = 4.8, 1.9, 0.9 Hz, 2H), 7.64 (td, *J* = 7.7, 1.9 Hz, 2H), 7.58 (d, *J* = 5.6 Hz, 1H), 7.39 (d, *J* = 1.2 Hz, 1H), 7.27 – 7.23 (m, 2H), 7.18 (dt, *J* = 8.0, 1.1 Hz, 2H), 7.03 (s, 1H), 7.02 (d, *J* = 3.7 Hz, 1H), 6.83 (s, 1H), 6.64 – 6.60 (m, 2H), 6.57 (d, *J* = 3.6 Hz, 1H), 4.05 (s, 3H), 4.02 (s, 3H), 3.20 (q, *J* = 7.3 Hz, 2H), 2.55 (s, 3H), 1.40 (t, *J* = 7.3 Hz, 3H). LCMS (m/z) $[M + H]^+$ = 567.1.

N-(1-(1,1-di(pyridin-2-yl)propyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**62d**). This compound was obtained by a procedure similar to the preparation of **62c** (96 mg, yield 79%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.64 (s, 2H), 7.72 – 7.48 (m, 5H), 7.38 (s, 1H), 7.25 – 7.13 (m, 3H), 7.04 (s, 1H), 6.80 (s, 1H), 6.56 (d, *J* = 15.4 Hz, 3H), 4.03 (d, *J* = 13.9 Hz, 6H), 3.17 (dq, *J* = 15.5, 7.4 Hz, 4H), 1.42 – 1.19 (m, 6H). LCMS (m/z) $[M + H]^+$ = 581.1.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)isobutyramide (**62e**). This compound was obtained by a procedure similar to the preparation of **62c** (78 mg, yield 82%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 8.88 (s, 1H), 8.65 (ddd, *J* = 4.8, 1.9, 0.9 Hz, 2H), 8.17 (s, 1H), 7.75 (td, *J* = 7.8, 1.9 Hz, 2H), 7.57 (d, *J* = 5.7 Hz, 1H), 7.35 (ddd, *J* = 7.6, 4.7, 1.1 Hz, 2H), 7.26 (d, *J* = 1.0 Hz, 1H), 7.25 – 7.22 (m, 2H), 7.00 (d, *J* = 3.5 Hz, 1H), 6.78 – 6.73 (m, 2H), 6.57 (s, 1H), 4.04 (s, 3H), 4.01 (s, 3H), 2.84 (hept, *J* = 6.8 Hz, 1H), 2.53 (s, 3H), 1.22 (d, *J* = 6.8 Hz, 6H). LCMS (m/z) $[M + H]^+$ = 545.2.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)acetamide (**62f**). This compound was obtained by a procedure similar to the

preparation of **62c** (98 mg, yield 81%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.69 (d, *J* = 3.6 Hz, 2H), 7.96 (s, 1H), 7.62 (td, *J* = 7.8, 1.9 Hz, 2H), 7.56 (d, *J* = 5.8 Hz, 1H), 7.44 (s, 1H), 7.28 – 7.20 (m, 2H), 7.16 (d, *J* = 8.0 Hz, 2H), 7.03 (d, *J* = 3.4 Hz, 2H), 6.64 (d, *J* = 5.8 Hz, 1H), 6.58 (s, 1H), 6.48 (d, *J* = 3.4 Hz, 1H), 4.04 (s, 3H), 4.00 (s, 3H), 2.55 (s, 3H), 2.28 (s, 3H). LCMS (m/z) [M + H]$^+$ = 517.2.

N-(1-benzhydryl-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (**25**). This compound was obtained by a procedure similar to the preparation of **27** (33 mg, yield 94%). ^{1}H NMR (400 MHz, DMSO-d_6) δ 10.93 (d, *J* = 5.6 Hz, 1H), 9.68 (s, 1H), 7.47 (s, 1H), 7.43 – 7.30 (m, 7H), 7.24 (d, *J* = 1.2 Hz, 1H), 7.22 – 7.15 (m, 5H), 6.97 (d, *J* = 3.3 Hz, 1H), 6.87 – 6.80 (m, 2H), 6.36 (dd, *J* = 7.0, 1.1 Hz, 1H), 4.09 (s, 3H), 3.09 (q, *J* = 7.3 Hz, 2H), 1.22 (t, *J* = 7.3 Hz, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 155.9, 139.9, 137.7, 130.2, 129.1, 128.8, 128.3, 128.2, 127.9, 126.6, 125.2, 123.4, 120.7, 117.3, 111.6, 105.6, 99.6, 99.4, 62.6, 45.5, 35.4, 8.2. HRMS (ESI) calcd for $C_{31}H_{29}N_4O_3S$ [M + H]$^+$, 537.1955; found, 537.1942.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)methanesulfonamide (26). This compound was obtained by a procedure similar to the preparation of **27** (56 mg, yield 92%). ^{1}H NMR (400 MHz, Methanol-d_4) δ 8.60 (d, *J* = 4.1 Hz, 2H), 7.77 – 7.67 (m, 2H), 7.34 (dd, *J* = 7.6, 4.9 Hz, 2H), 7.23 (s, 1H), 7.16 (d, *J* = 4.2 Hz, 2H), 7.14 (s, 2H), 6.82 (d, *J* = 7.0 Hz, 1H), 6.79 (d, *J* = 3.5 Hz, 1H), 6.42 (s, 1H), 6.16 (d, *J* = 6.9 Hz, 1H), 4.08 (s, 3H), 2.98 (s, 3H), 2.51 (s, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.1, 155.8, 148.9, 137.1, 137.0, 130.5, 128.9, 127.9, 127.5, 127.4, 125.1, 123.2, 122.8, 122.5, 122.2, 117.2, 111.9, 108.2, 99.5, 99.1, 69.9, 35.3, 30.7, 27.9. HRMS (ESI) calcd for $C_{29}H_{27}FN_6O_3S$ [M + H]$^+$, 539.1860; found, 539.1836.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (27). A mixture of **62c** (5.66 g, 10 mmol), TMSCl (2.03 mL, 16 mmol), NaI (2.40 g, 16 mmol), and H_2O (90 μL, 5 mmol) in acetonitrile (30

mL) was heated at 50 °C for 3 h, and then cooled to room temperature. The mixture was then diluted with ethyl acetate and washed three times with aqueous $NaHCO_3$. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was recrystallized (methanol/ethyl acetate) to obtain the final product **27** (4.7 g, yield 85%). ^{1}H NMR (400 MHz, DMSO-d_6) δ 10.92 (d, *J* = 5.4 Hz, 1H), 9.66 (s, 1H), 8.63 (d, *J* = 3.9 Hz, 2H), 7.79 (td, *J* = 7.8, 1.9 Hz, 2H), 7.38 (dd, *J* = 7.5, 4.8 Hz, 2H), 7.22 (s, 1H), 7.16 (d, *J* = 8.0 Hz, 2H), 7.12 (s, 1H), 7.10 (d, *J* = 3.4 Hz, 1H), 6.87 (d, *J* = 3.4 Hz, 1H), 6.77 (t, *J* = 6.3 Hz, 1H), 6.32 (s, 1H), 5.78 (d, *J* = 7.0 Hz, 1H), 4.03 (s, 3H), 3.12 (q, *J* = 7.3 Hz, 2H), 2.43 (s, 3H), 1.25 (t, *J* = 7.3 Hz, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.1, 155.8, 148.9, 137.1, 137.0, 130.3, 128.9, 127.9, 127.5, 127.4, 125.1, 123.2, 122.8, 122.5, 121.8, 117.2, 111.4, 108.2, 99.4, 99.1, 69.9, 45.6, 35.3, 27.9, 8.2. HRMS (ESI) calcd for $C_{30}H_{29}FN_6O_3S$ $[M + H]^+$, 553.2016; found, 553.2017.

N-(1-(1,1-di(pyridin-2-yl)propyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)ethanesulfonamide (28). This compound was obtained by a procedure similar to the preparation of **27** (62 mg, yield 91%). ^{1}H NMR (400 MHz, Methanol-d_4) δ 8.93 (dd, *J* = 5.4, 1.7 Hz, 2H), 8.13 (td, *J* = 8.0, 1.7 Hz, 2H), 8.07 (s, 1H), 7.95 (d, *J* = 3.6 Hz, 1H), 7.75 (dd, *J* = 7.6, 5.3 Hz, 2H), 7.25 (d, *J* = 1.2 Hz, 1H), 7.21 (d, *J* = 8.2 Hz, 2H), 7.09 (d, *J* = 3.5 Hz, 1H), 7.05 (s, 1H), 6.80 (d, *J* = 7.0 Hz, 1H), 6.05 (s, 1H), 5.76 (d, *J* = 7.0 Hz, 1H), 4.03 (s, 3H), 3.17 (q, *J* = 7.4 Hz, 2H), 2.96 (q, *J* = 7.3 Hz, 2H), 1.37 (t, *J* = 7.4 Hz, 3H), 0.84 (t, *J* = 7.2 Hz, 3H). ^{13}C NMR (101 MHz, Methanol-d_4) δ 164.4, 160.5, 158.0, 146.0, 144.0, 138.1, 132.2, 131.24, 131.17, 130.7, 128.0, 126.3, 125.8, 125.7, 124.6, 124.0, 118.9, 114.6, 109.0, 102.2, 102.0, 68.5, 47.2, 38.4, 36.1, 9.0, 8.6. HRMS (ESI) calcd for $C_{31}H_{31}N_6O_3S$ $[M + H]^+$, 567.2173; found, 567.2151.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)isobutyramide (**29**). This compound was obtained by a procedure similar to the preparation of **27** (39 mg, yield 93%). ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.11 (s, 1H), 8.89 (s, 1H), 8.64 (ddd, *J* = 4.7, 1.9, 0.9 Hz, 2H), 8.10 (s, 1H), 7.76 (td, *J* = 7.8, 1.9 Hz, 2H),

7.35 (ddd, *J* = 7.6, 4.7, 1.1 Hz, 2H), 7.23 (dt, *J* = 8.1, 1.1 Hz, 2H), 7.12 (s, 1H), 7.00 (d, *J* = 3.5 Hz, 1H), 6.87 (d, *J* = 7.1 Hz, 1H), 6.76 (dd, *J* = 3.5, 0.9 Hz, 1H), 6.47 (t, *J* = 1.1 Hz, 1H), 6.08 (d, *J* = 7.1 Hz, 1H), 4.13 (s, 3H), 2.82 (m, 1H), 2.53 (s, 3H), 1.22 (d, *J* = 6.8 Hz, 6H). ^{13}C NMR (101 MHz, Acetone-d_6) δ 176.2, 162.7, 157.1, 149.9, 138.1, 137.5, 132.6, 132.0, 129.5, 128.9, 127.6, 125.5, 124.6, 123.8, 123.5, 121.4, 119.8, 111.8, 108.8, 101.0, 98.9, 71.2, 36.4, 35.7, 27.9, 20.2. HRMS (ESI) calcd for $C_{32}H_{30}N_6O_2$ [M + H]$^+$, 531.2503; found, 531.2489.

N-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(1-methyl-7-oxo-6,7-dihydro-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indol-4-yl)acetamide (**30**). This compound was obtained by a procedure similar to the preparation of **27** (44 mg, yield 96%). ^{1}H NMR (400 MHz, Chloroform-*d*) δ 9.17 (s, 1H), 8.74 – 8.63 (m, 2H), 7.93 (s, 1H), 7.63 (t, *J* = 7.8 Hz, 2H), 7.40 (s, 1H), 7.25 – 7.20 (m, 2H), 7.14 (d, *J* = 8.1 Hz, 2H), 7.03 (d, *J* = 3.4 Hz, 1H), 6.95 (s, 1H), 6.71 (t, *J* = 6.1 Hz, 1H), 6.47 (s, 1H), 6.15 (d, *J* = 7.0 Hz, 1H), 4.14 (s, 3H), 2.56 (s, 3H), 2.29 (s, 3H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 168.6, 161.2, 155.8, 148.9, 136.9, 136.8, 131.3, 128.7, 127.9, 127.2, 126.7, 125.0, 123.1, 122.8, 122.5, 120.3, 117.7, 110.8, 107.3, 99.3, 99.1, 69.8, 35.2, 27.9, 23.9. HRMS (ESI) calcd for $C_{30}H_{27}N_6O_2$ [M + H]$^+$, 503.2190; found, 503.2170.

Methyl 6-bromo-1-(di(pyridin-2-yl)methyl)-1*H*-indole-4-carboxylate (**64**). To a solution of **63** (1.27 g, 5 mmol) in DMF (20 mL) at RT was added K_2CO_3 (1.38 g, 10 mmol) and 2,2'-(bromomethylene)dipyridine (1.87 g, 7.5 mmol) and then the mixture was stirred at 50 °C, and quenched after 16 h by addition of water. The solution was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 2:1) to obtain **64** (1.52 g, yield 72%) as a yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.55 – 8.52 (m, 2H), 7.95 (d, *J* = 1.7 Hz, 1H), 7.70 (t, *J* = 1.2 Hz, 1H), 7.54 (td, *J* = 7.7, 1.8 Hz, 2H), 7.32 (d, *J* = 3.3 Hz, 1H), 7.16 – 7.09 (m, 3H), 7.06 – 7.01 (m, 2H), 6.93 (s, 1H), 3.87 (s, 3H). LCMS (m/z) [M + H]$^+$ = 422.0.

Methyl 6-bromo-1-(1,1-di(pyridin-2-yl)ethyl)-1*H*-indole-4-carboxylate (**65**). To a solution of **64** (1.27 g, 3 mmol) in THF (10 mL) at 0 °C was added *t*-BuOK (1.01 g, 9 mmol) and MeI (0.56 mL, 9 mmol) successively. Then the solution was stirred at RT for 2 h and water was added to quench the reaction. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Filtration and concentration in vacuo gave a residue, which was purified by flash column chromatography on silica gel (Hexanes-EtOAc, 3:1) to obtain **65** (1.15 g, yield 88%) as a yellow solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.63 (ddd, *J* = 4.8, 1.8, 0.9 Hz, 2H), 8.00 (d, *J* = 1.7 Hz, 1H), 7.70 (s, 1H), 7.67 (td, *J* = 7.8, 1.9 Hz, 2H), 7.32 (d, *J* = 3.3 Hz, 1H), 7.27 (d, *J* = 1.1 Hz, 1H), 7.28 – 7.21 (m, 1H), 7.16 (dd, *J* = 3.3, 0.8 Hz, 1H), 7.10 (d, *J* = 1.1 Hz, 1H), 7.08 (d, *J* = 1.0 Hz, 1H), 6.95 (s, 1H), 3.97 (s, 3H). LCMS (m/z) $[M + H]^+$ = 436.0.

Methyl 1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1*H*-pyrrolo[2,3-*c*]pyridin-3-yl)-1*H*-indole-4-carboxylate (**66**). A mixture of **65** (870 mg, 2 mmol), Pd(dppf)Cl_2-DCM (163 mg, 0.2 mmol), **35** (693 mg, 2.4 mmol), and K_3PO_4 (1.06 g, 5 mmol) in dioxane-water (10 mL, 4:1) was heated at 90 °C for 24 h. The mixture was then diluted with ethyl acetate and filtered to obtain a clear solution, which was washed three times with brine. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product was purified by flash column chromatography on silica gel (Hexanes-Acetone, 1:1) to yield **66** (880 mg, yield 85%) as a grey solid. ^{1}H NMR (400 MHz, Chloroform-*d*) δ 8.69 (ddd, *J* = 4.8, 1.9, 0.9 Hz, 2H), 8.02 (d, *J* = 1.4 Hz, 1H), 7.63 (td, *J* = 7.8, 1.9 Hz, 2H), 7.58 (d, *J* = 5.8 Hz, 1H), 7.25 (dd, *J* = 4.9, 1.2 Hz, 1H), 7.23 (dd, *J* = 4.8, 1.1 Hz, 1H), 7.21 – 7.16 (m, 4H), 7.05 (s, 1H), 7.01 (t, *J* = 1.1 Hz, 1H), 6.59 (d, *J* = 5.8 Hz, 1H), 4.06 (s, 3H), 4.03 (s, 3H), 4.00 (s, 3H), 2.57 (s, 3H). LCMS (m/z) $[M + H]^+$ = 518.3.

2-(1-(1,1-di(pyridin-2-yl)ethyl)-6-(7-methoxy-1-methyl-1H-pyrrolo[2,3-c]pyridin-3-yl)-1H-indol-4-yl)propan-2-ol (**67**). A mixture of **66** (776 mg, 1.5 mmol), TMSCl (0.31 mL, 2.4 mmol), NaI (360 mg, 2.4 mmol), and H_2O (15 μL, 0.8 mmol) in acetonitrile (5 mL) was heated at 50 °C

for 3 h, and then cooled to room temperature. The mixture was then diluted with ethyl acetate and washed three times with aqueous $NaHCO_3$. The organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated. The crude product **67** (720 mg) was obtained without further purification. LCMS (m/z) $[M + H]^+$ = 504.2.

3-(1-(1,1-di(pyridin-2-yl)ethyl)-4-(2-hydroxypropan-2-yl)-1*H*-indol-6-yl)-1-methyl-1,6-dihydro-7*H*-pyrrolo[2,3-*c*]pyridin-7-one (31). To a solution of **67** (50 mg, 0.1 mmol) in THF (3 mL) at -42 °C was added MeLi (0.8 mmol) and the mixture was stirred for 2 h. Water was added to quench the reaction. The mixture was extracted with ethyl acetate and the combined organic layer was dried over anhydrous Na_2SO_4. Then filtration and concentration in vacuo gave a residue, which was purified by preparative TLC (DCM-MeOH, 20:1) to obtain **31** (27 mg, yield 55%) as a grey solid. ^{1}H NMR (400 MHz, Acetone-d_6) δ 10.06 (s, 1H), 8.65 (ddd, J = 4.8, 1.9, 0.9 Hz, 2H), 7.77 (td, J = 7.8, 1.9 Hz, 2H), 7.37 – 7.35 (m, 2H), 7.34 (dd, J = 4.8, 1.1 Hz, 1H), 7.28 (dt, J = 8.0, 1.0 Hz, 2H), 7.16 (s, 1H), 6.98 (d, J = 3.5 Hz, 1H), 6.85 (d, J = 7.3 Hz, 2H), 6.83 (dd, J = 3.5, 0.9 Hz, 1H), 6.65 (t, J = 1.2 Hz, 1H), 5.96 (d, J = 7.0 Hz, 1H), 4.11 (s, 3H), 2.53 (s, 3H), 1.73 (s, 6H). ^{13}C NMR (101 MHz, DMSO-d_6) δ 161.3, 155.8, 148.9, 142.6, 136.9, 136.7, 128.8, 128.0, 126.6, 126.3, 124.9, 124.7, 123.1, 122.7, 122.7, 118.0, 115.0, 109.8, 102.4, 99.4, 71.4, 69.7, 35.2, 31.1, 27.6. HRMS (ESI) calcd for $C_{31}H_{30}N_5O_2$ $[M + H]^+$, 504.2394; found, 504.2358.

Supporting Information

Synthesis and characterization of all compounds and further experimental details.

Molecular formula strings (CSV)

Accession Codes

The X-ray coordinates have been deposited with the Protein Data Bank. The codes are 6P05 (BRD4 BD1 domain with **27**). Authors will release the atomic coordinates and experimental data upon article publication.

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Author contributions:

The manuscript was written and the research directed by R.X. and G.T. Drug design and synthesis was directed by R.X. The largest contribution to synthesis was from Y.L. and R.X. Intermediate synthesis, scale-up, synthetic optimization, and purification was supported by Z.S. Cell line and bioassay development was led by J.Z. and R.X. Animal study was led by D.T. and J.Z. Biological support was provided by L.G. and F.H. PK was supported by D.K. and O.D. X-ray co-crystal structure was led by K.R. All authors have given approval to the final version of the manuscript.

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Abbreviations

ADMET, absorption, distribution, metabolism, excretion and toxicity; AI, aromatase inhibitors; BET, bromodomain and extra-terminal; CDK, cyclin-dependent kinase; CYP450, Cytochrome P450; DMF, dimethylformamide; E2, estradiol; EP300, E1A-associated protein p300; ER, estrogen receptor; ERE, estrogen response elements; ET, extra C-terminal domain; hERG, Ful, fulvestrant, The human ether-a-go-go-related gene; HPBCD, Hydroxypropyl beta cyclodextrin; LLE, Ligand-lipophilicity efficiency; MED1, Mediator Complex Subunit 1; nM,

nanomolar; PEG400, polyethylene glycol 400; PK, pharmacokinetics; PROTAC, proteolysis targeting chimera; P-TEFb, positive transcription elongation factor b; RNAPII, RNA polymerase II; SAR, structure–activity relationship; SERD, selective estrogen receptor degrader; TAF-1, TBP-associated factor 1; TAM, tamoxifen; TR-FRET, Time-Resolved Fluorescence Resonance Energy Transfer; YY1, Yin Yang 1; μM, micromolar.

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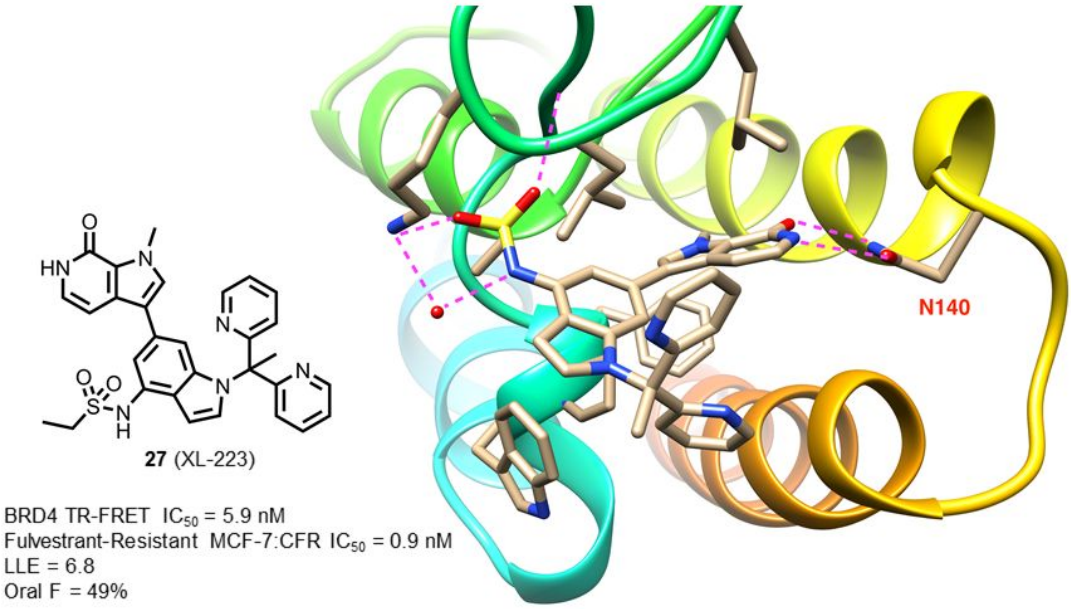

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